

Relational convolution: does overlap itself contain information?

Bounded real-PRISM and symmetric-synthetic feasibility experiment

$S(x, y, dx, dy)$

400 complete real local surfaces | 160,000 directed views | 80,200 canonical pairs

Relative Delta correlation

0.71

Absolute Delta correlation

0.94

Absolute gradient similarity

0.93

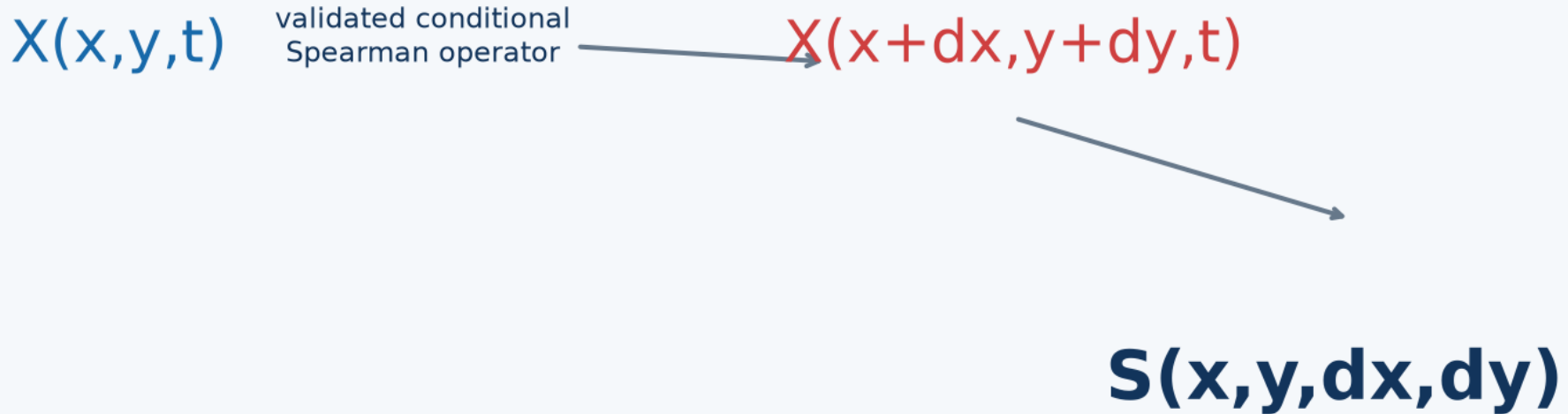
Station Delta correlation

0.68

Result: overlap is informative in this block, but it is not yet a production verb or validated climate-boundary product.

The object is a field of relationships

Cold and warm conditional synchrony remain separate; $\Delta S = S_{\text{cold}} - S_{\text{warm}}$



Relative coordinates (dx,dy) describe shape around the focal pixel. Absolute coordinates $q=(x+dx,y+dy)$ identify the fixed geography observed by overlapping surfaces. The experiment keeps cold, warm, Delta, valid counts, focal/comparison coordinates, distance, bearing, and canonical pair identity before any reduction.

Why convolution-like - and why not a CNN

The local operation translates; the relationship is estimated rather than multiplied by fixed weights

Ordinary convolution

$$Y(p) = \sum K(\delta) X(p+\delta)$$

fixed weights
immediate sum
one output

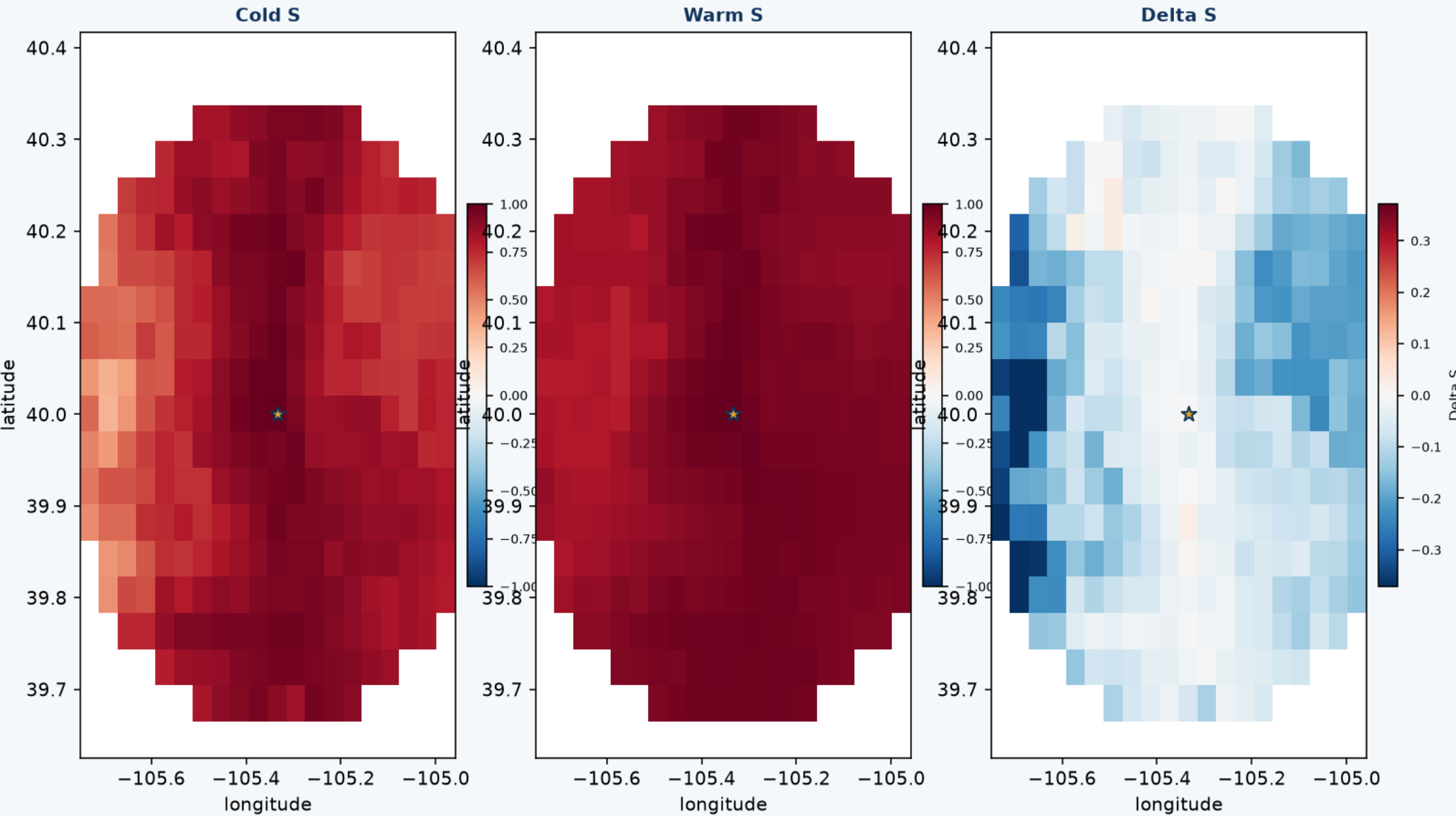
Relational surface

$$S_p(\delta) = f(X_p(t), X_{(p+\delta)}(t))$$

estimated pair relation
complete neighborhood retained
no training objective

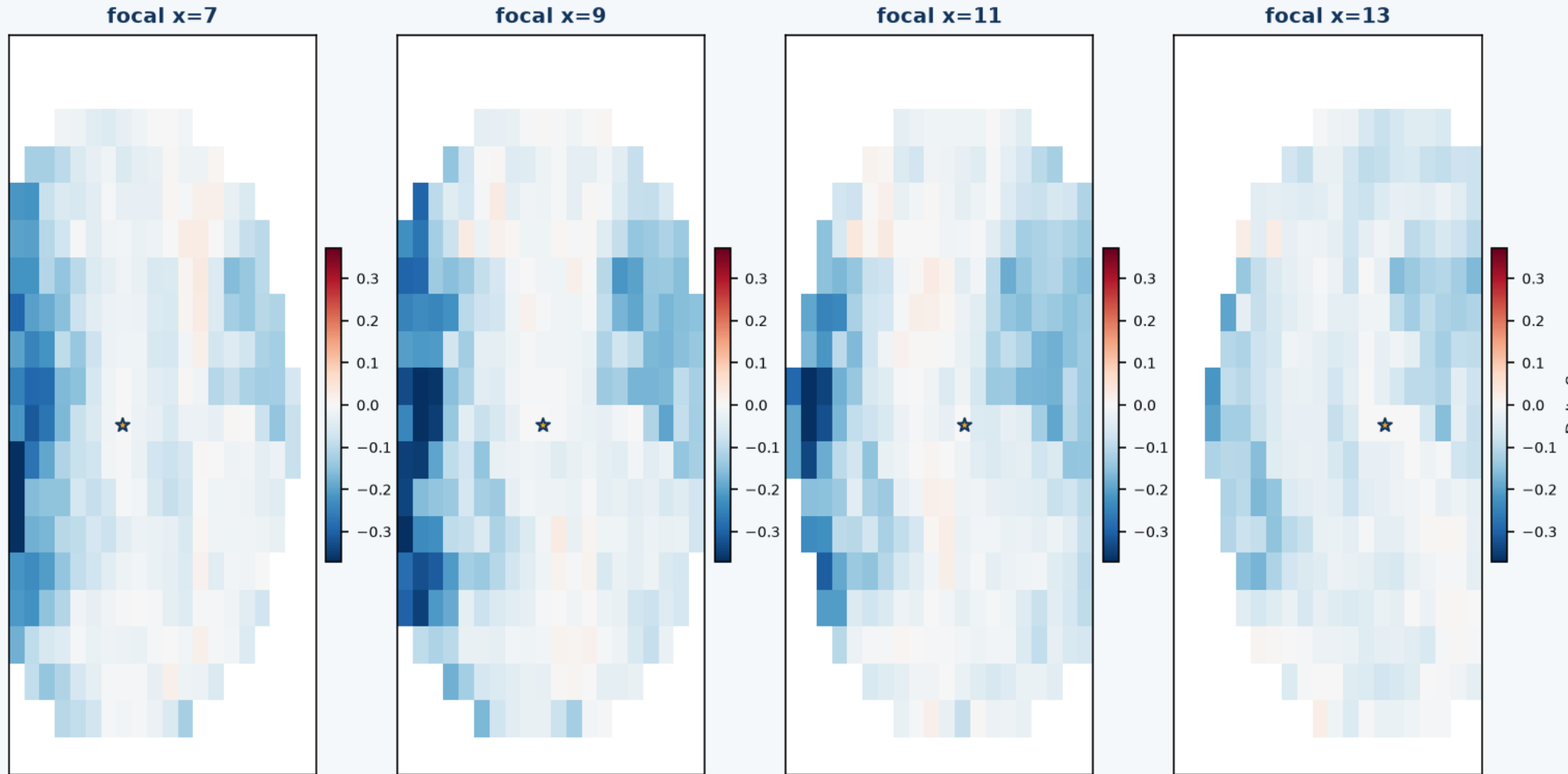
One real local synchrony surface

Front Range focal pixel (-105.333, 40.000); 40 km observation window



Neighboring focal surfaces are not independent pictures

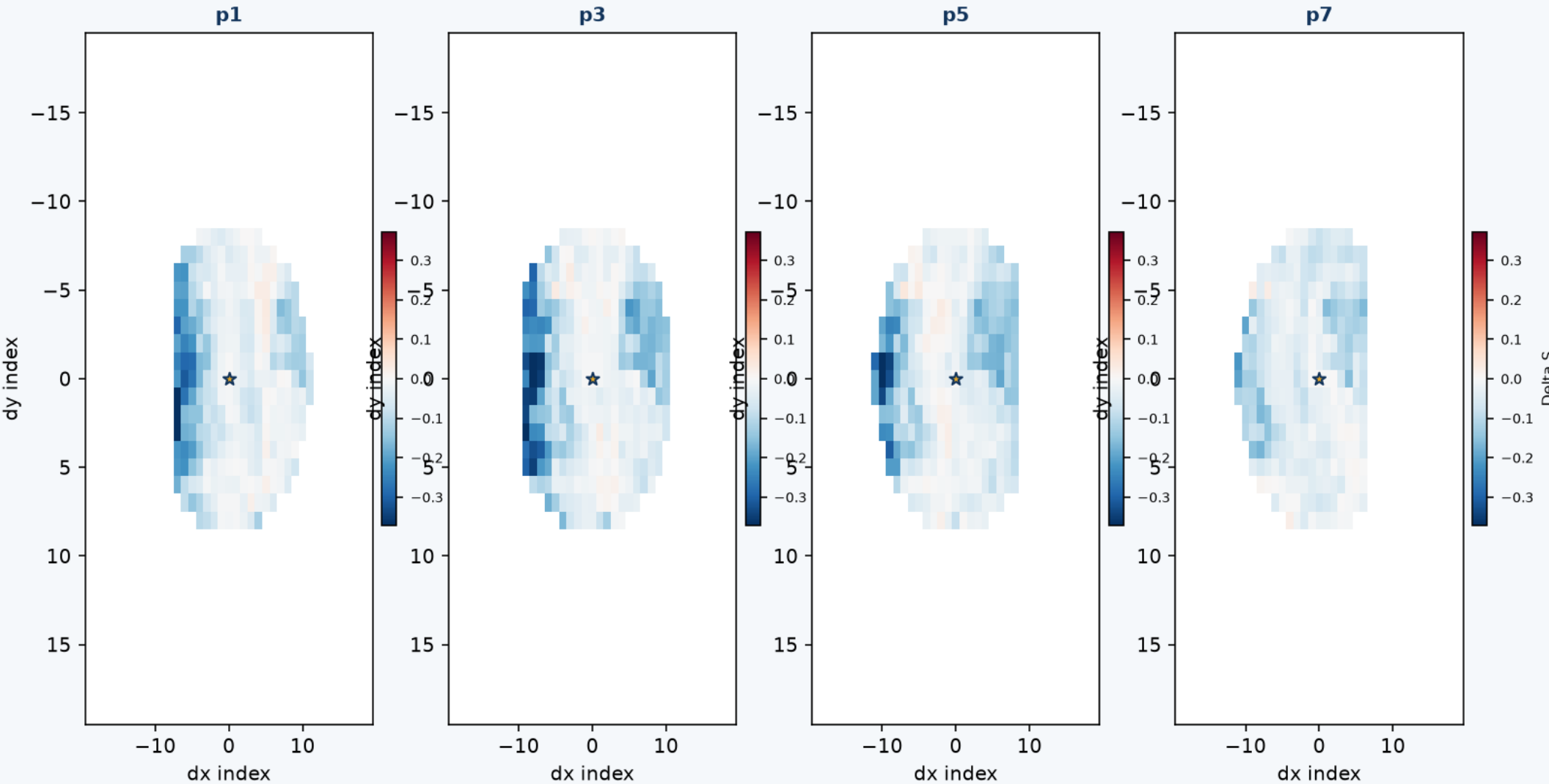
A west-east transect: the same geographic cells are revisited from moving focal locations



The panels share absolute map coordinates. Surface values change with focal location, but mountains, plains, and any fixed transition stay in place.

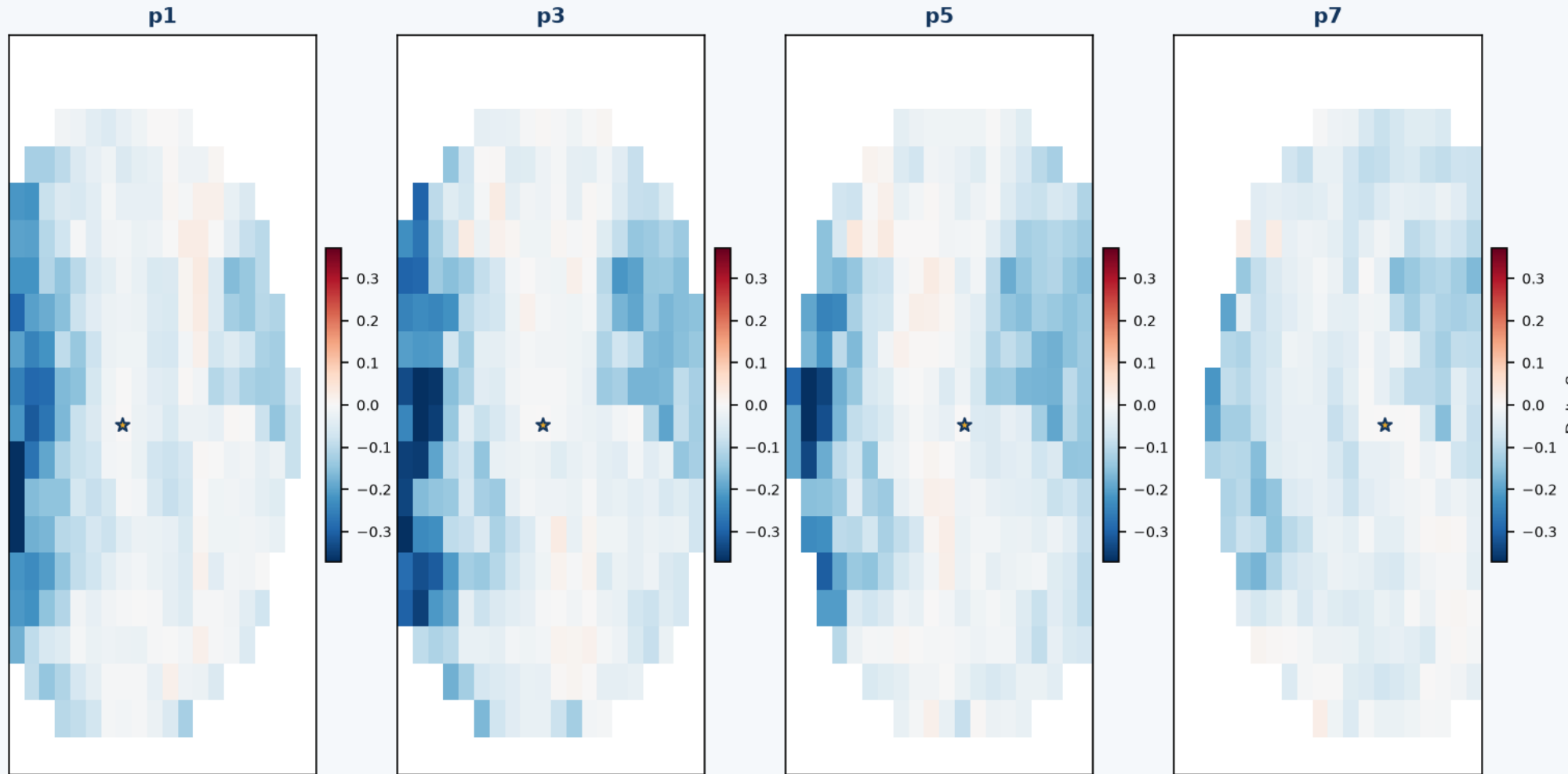
Relative-coordinate view: geography moves through the window

Every focal is recentered at (0,0); equal offsets do not refer to equal absolute locations



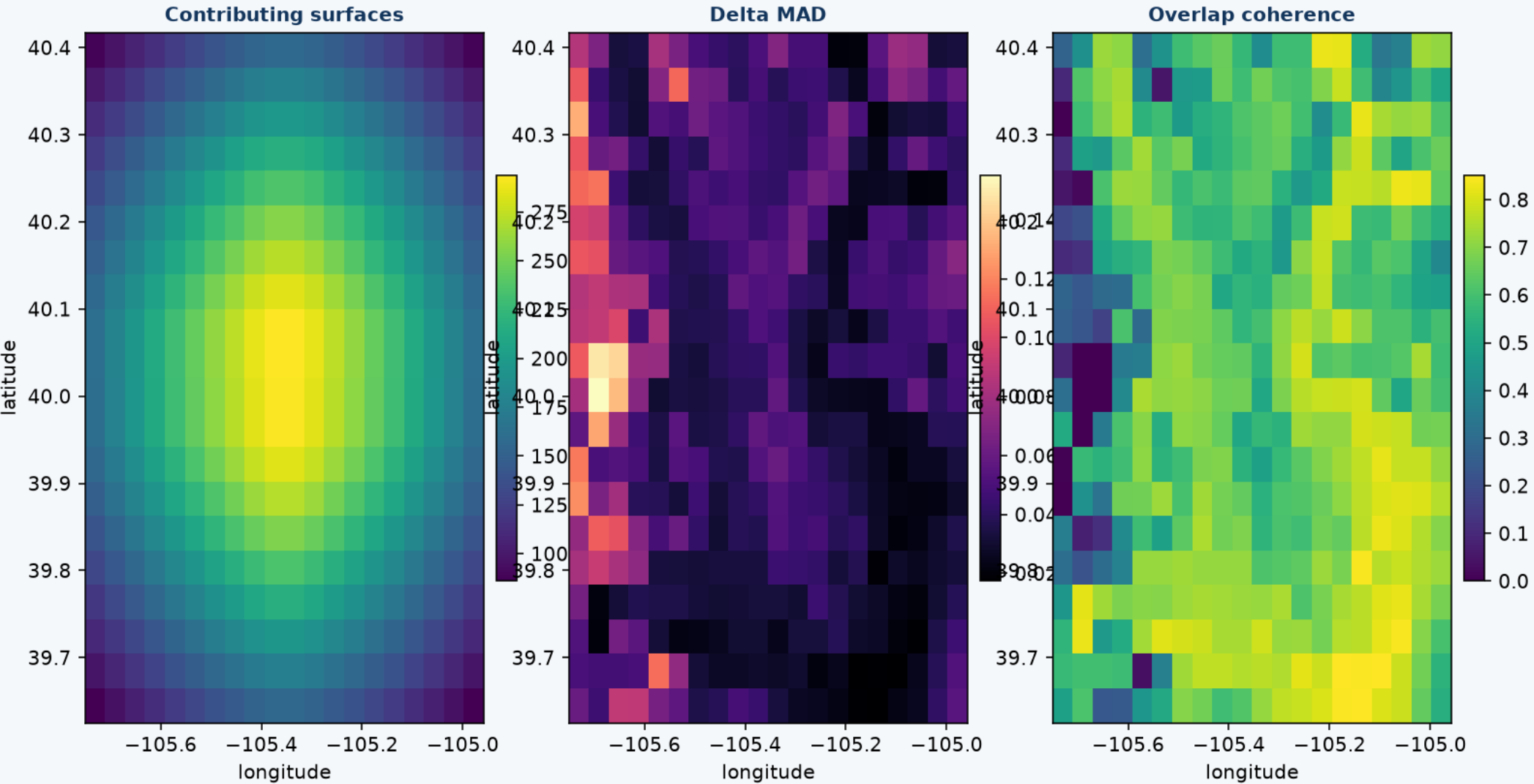
Absolute-geography view: known alignment, no learned registration

The same four surfaces are mapped to comparison latitude/longitude



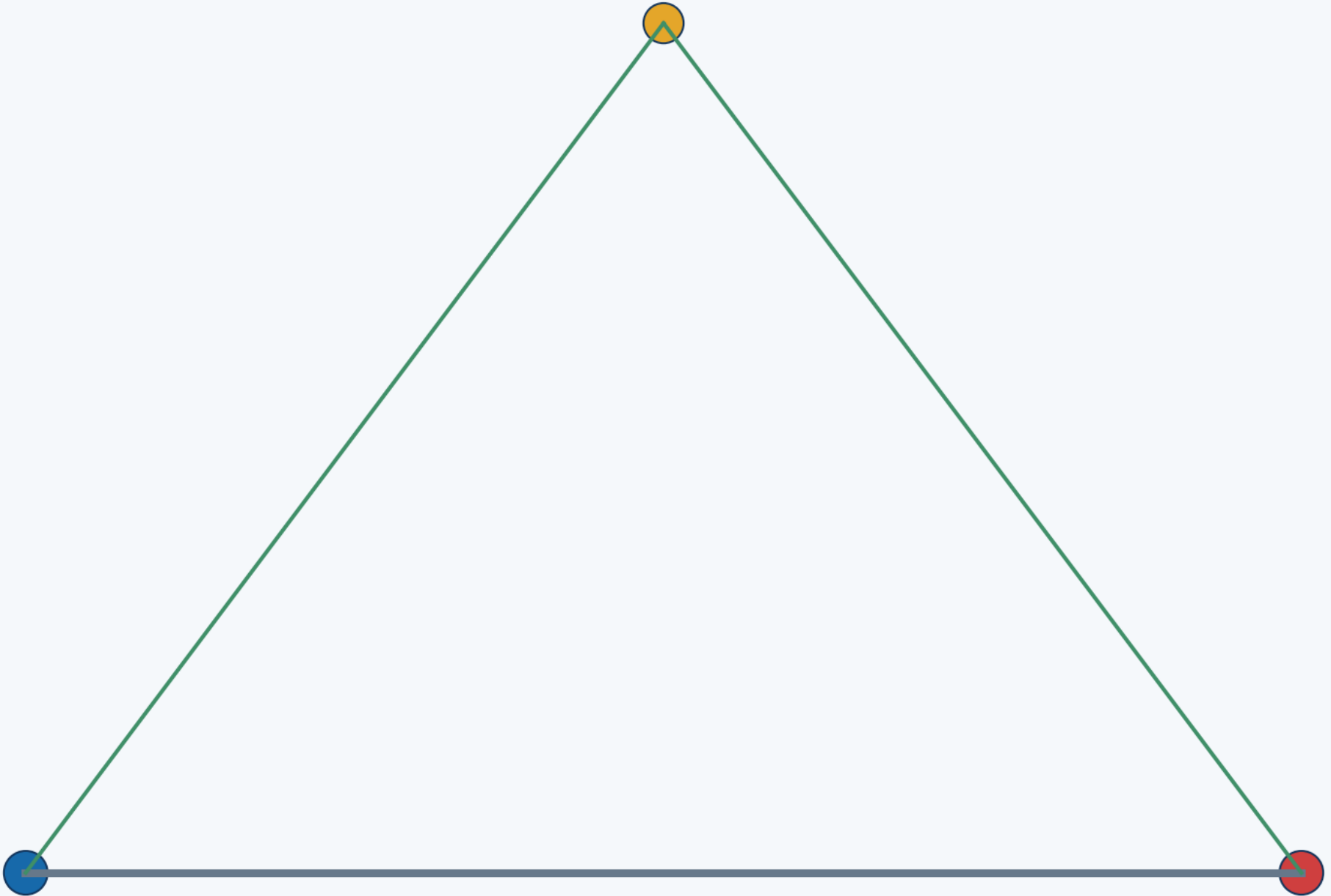
The explicit overlap-support object

Raw contributors remain in `real_complete_surfaces.nc`; these maps are inspectable projections



Repeated geography is not the same as repeated pairs

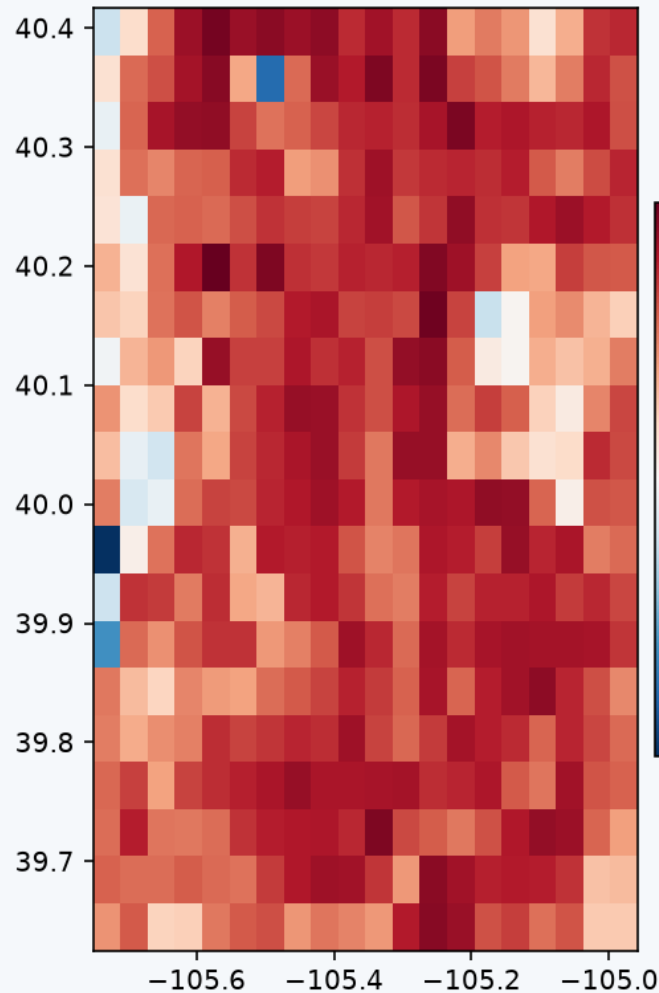
Canonical-pair identity prevents endpoint reversal from masquerading as replicate evidence



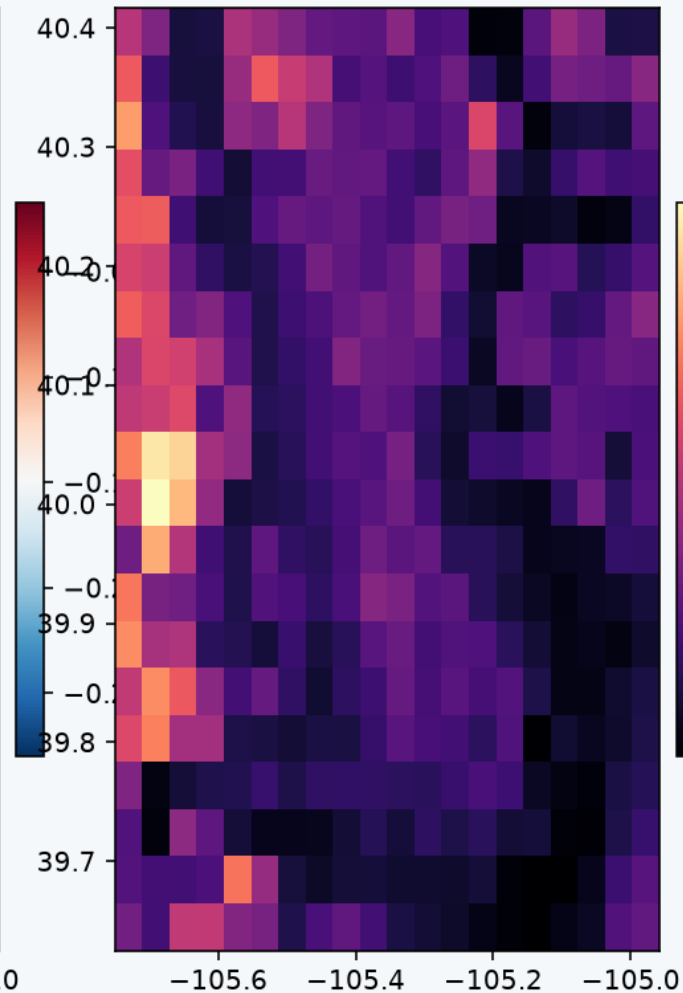
Median and dispersion retain magnitude, not 2-D geometry

Real 40 km support; independent surface summaries

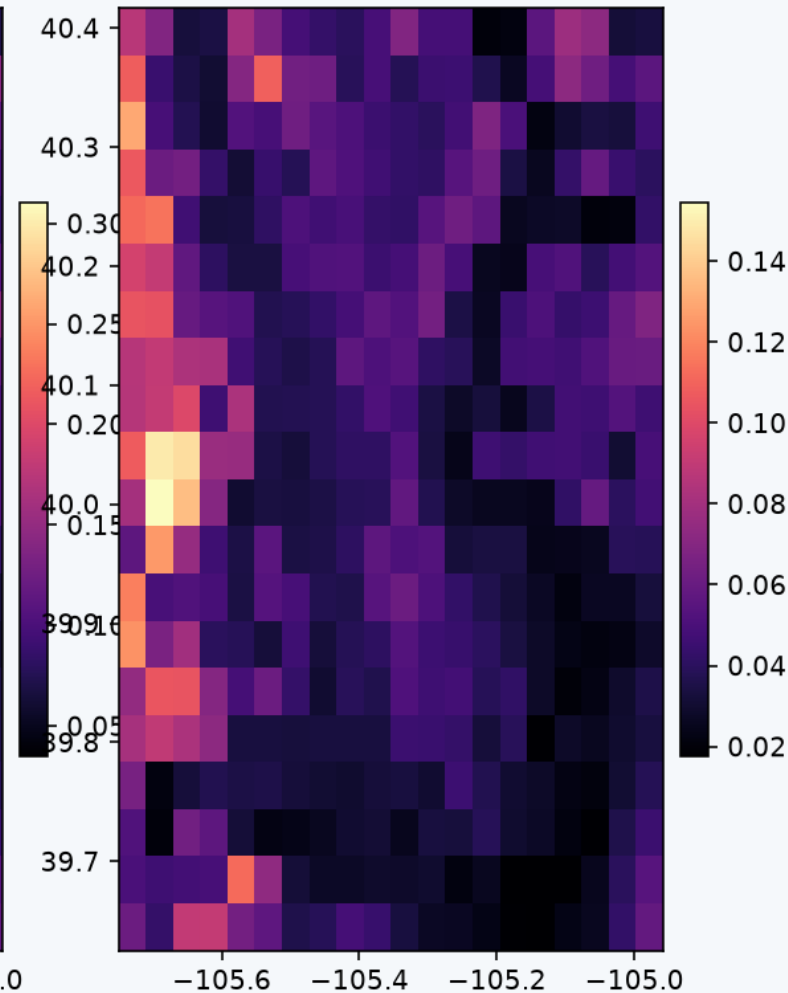
Median Delta S



IQR



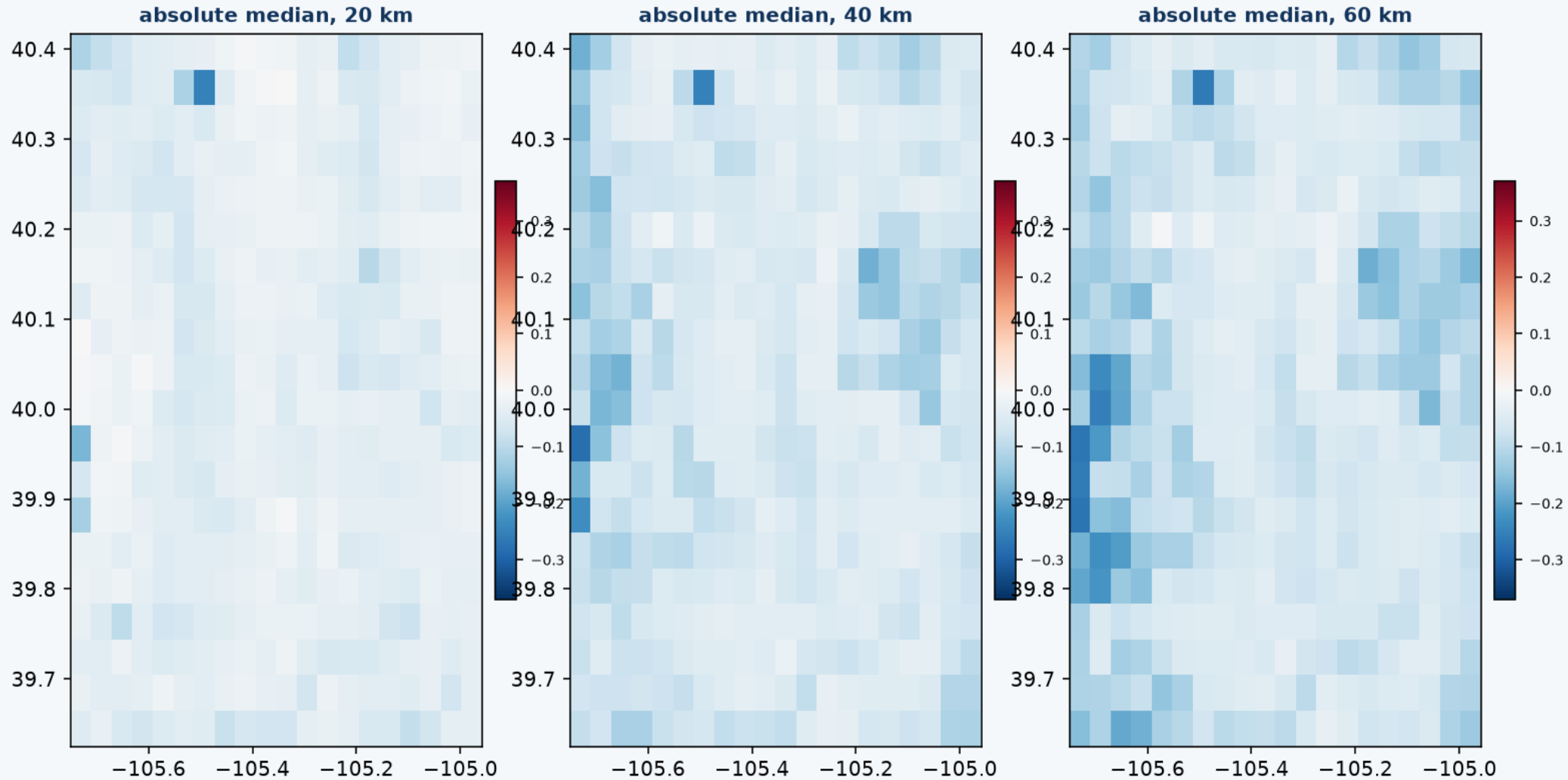
MAD



Median-only normalized reconstruction RMSE = 1.05; IQR/MAD describe spread but cannot restore where values occur.

Observation-window sensitivity is not radius selection

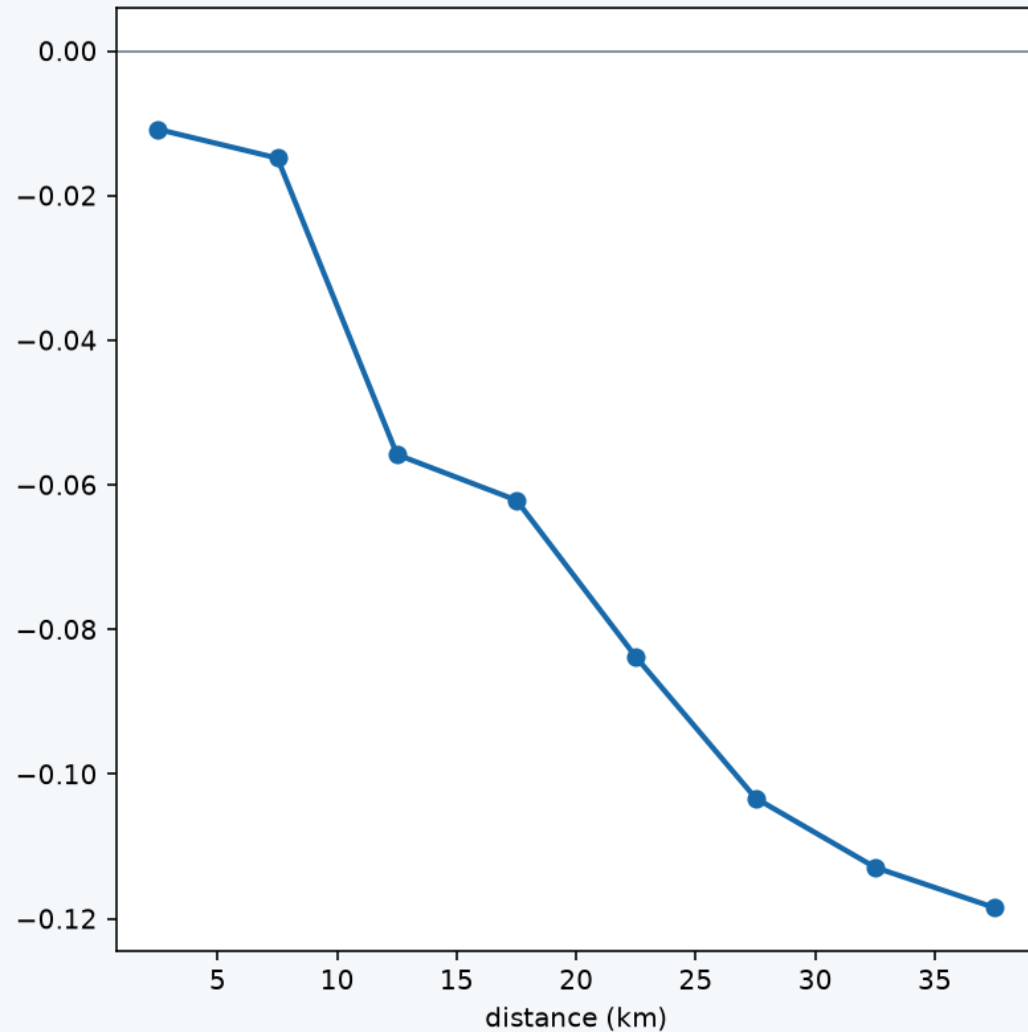
20, 40, and 60 km are computational windows; 60 km is edge-censored in this block



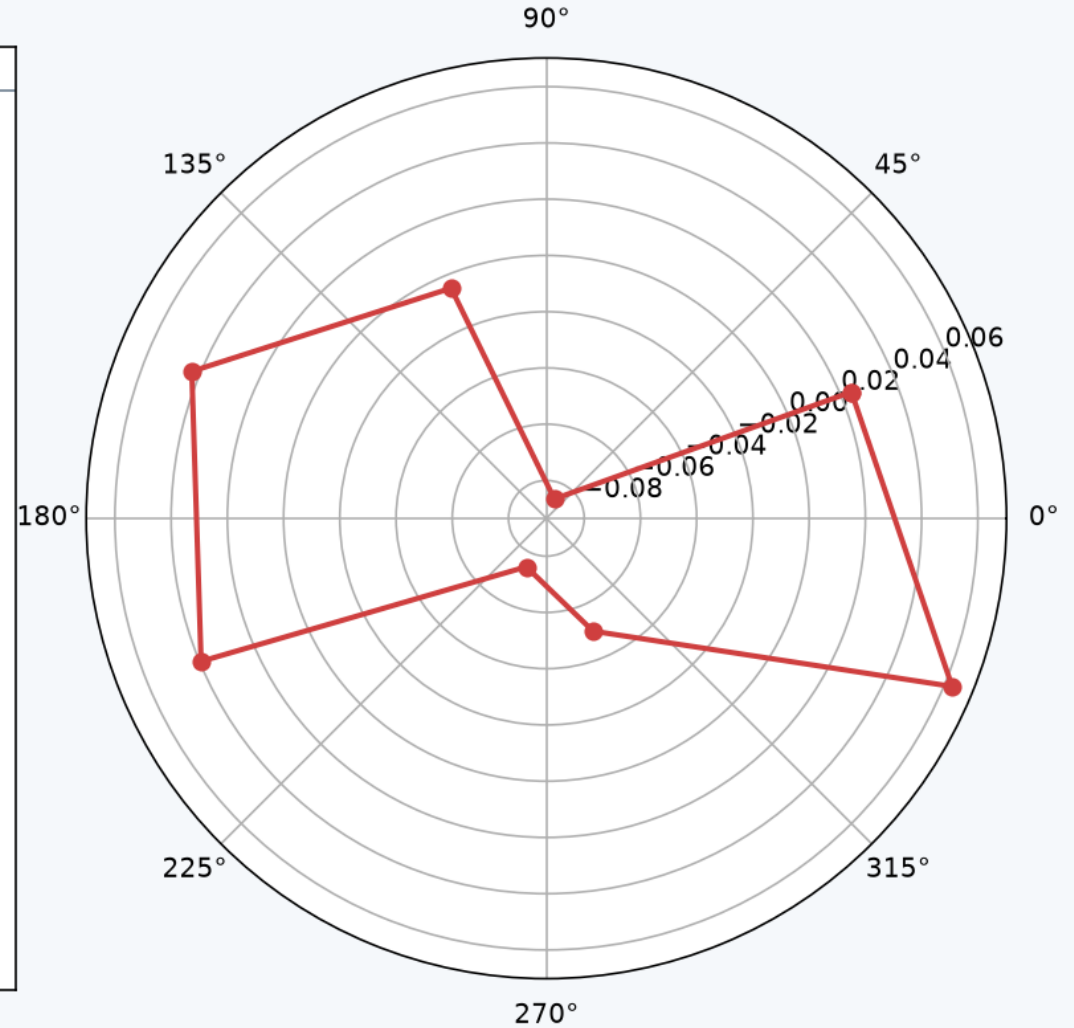
Radial plus directional summaries are the strongest independent baseline

Fine empirical strata; no monotonic decay is imposed

Fine radial profile



Radially adjusted direction

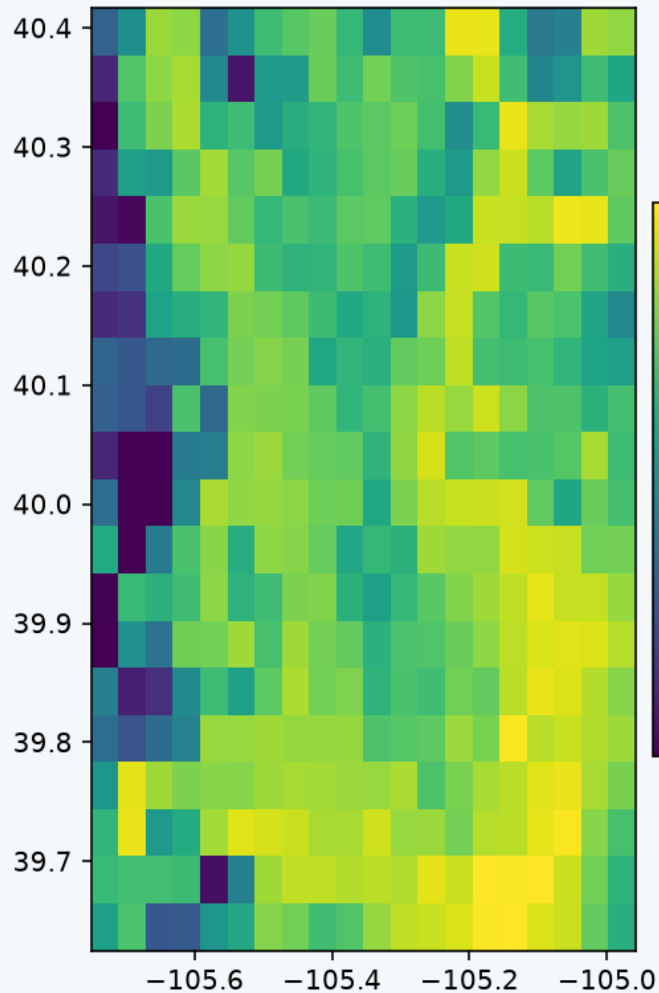


Radial + directional nRMSE = 0.68 versus median-only 1.05.

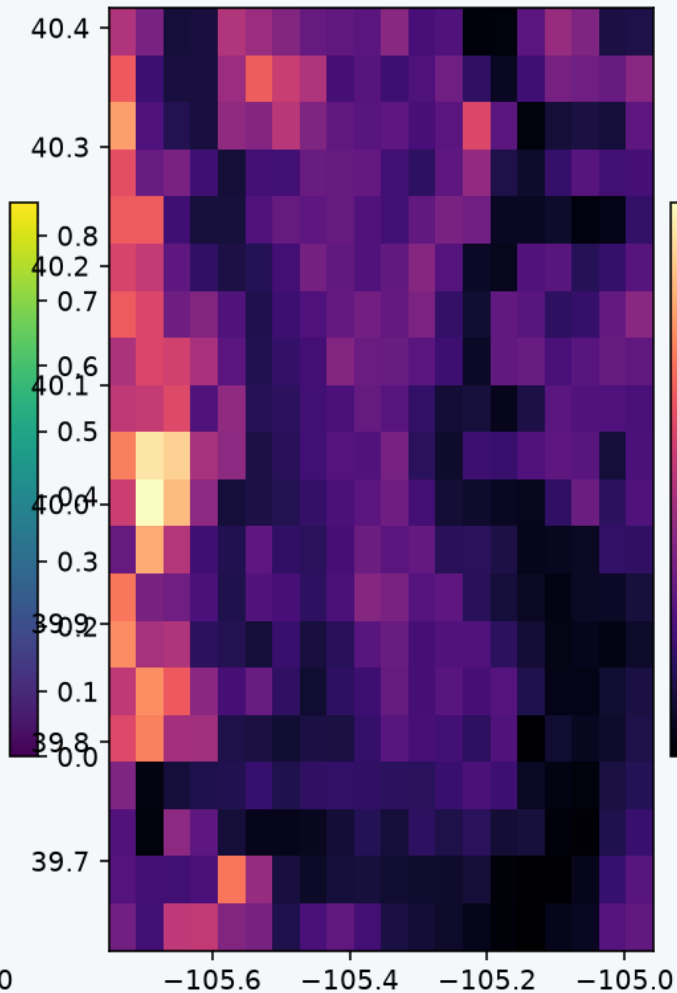
Candidate A - overlap coherence

Agreement and dispersion at the same absolute locations; contributor count stays visible

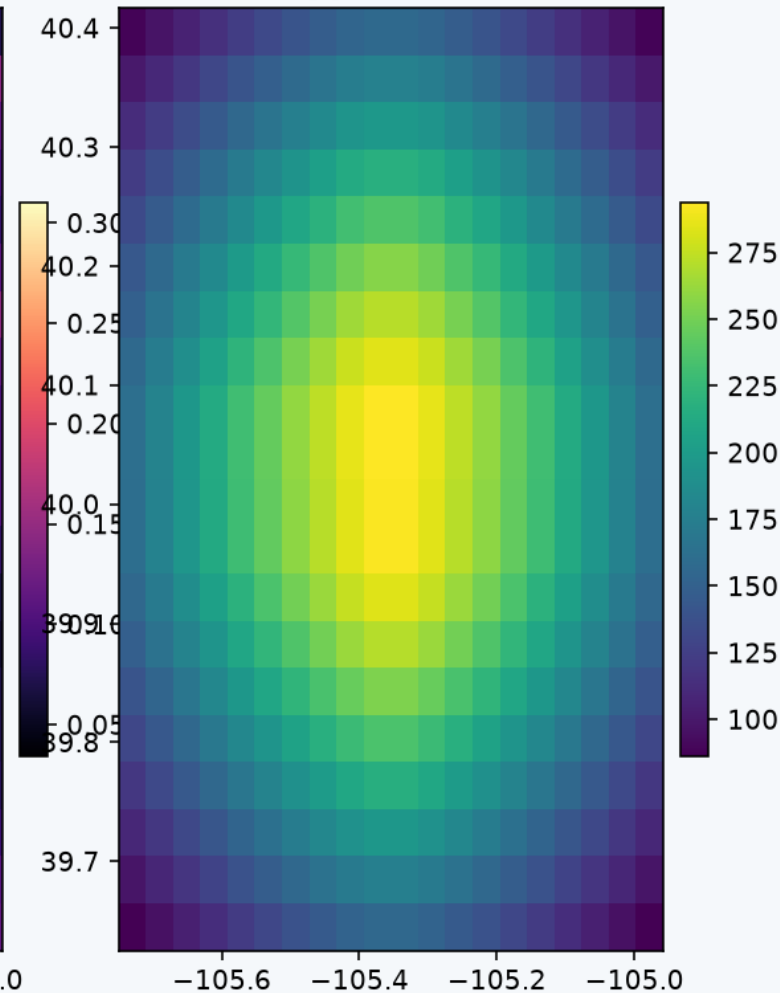
coherence



IQR

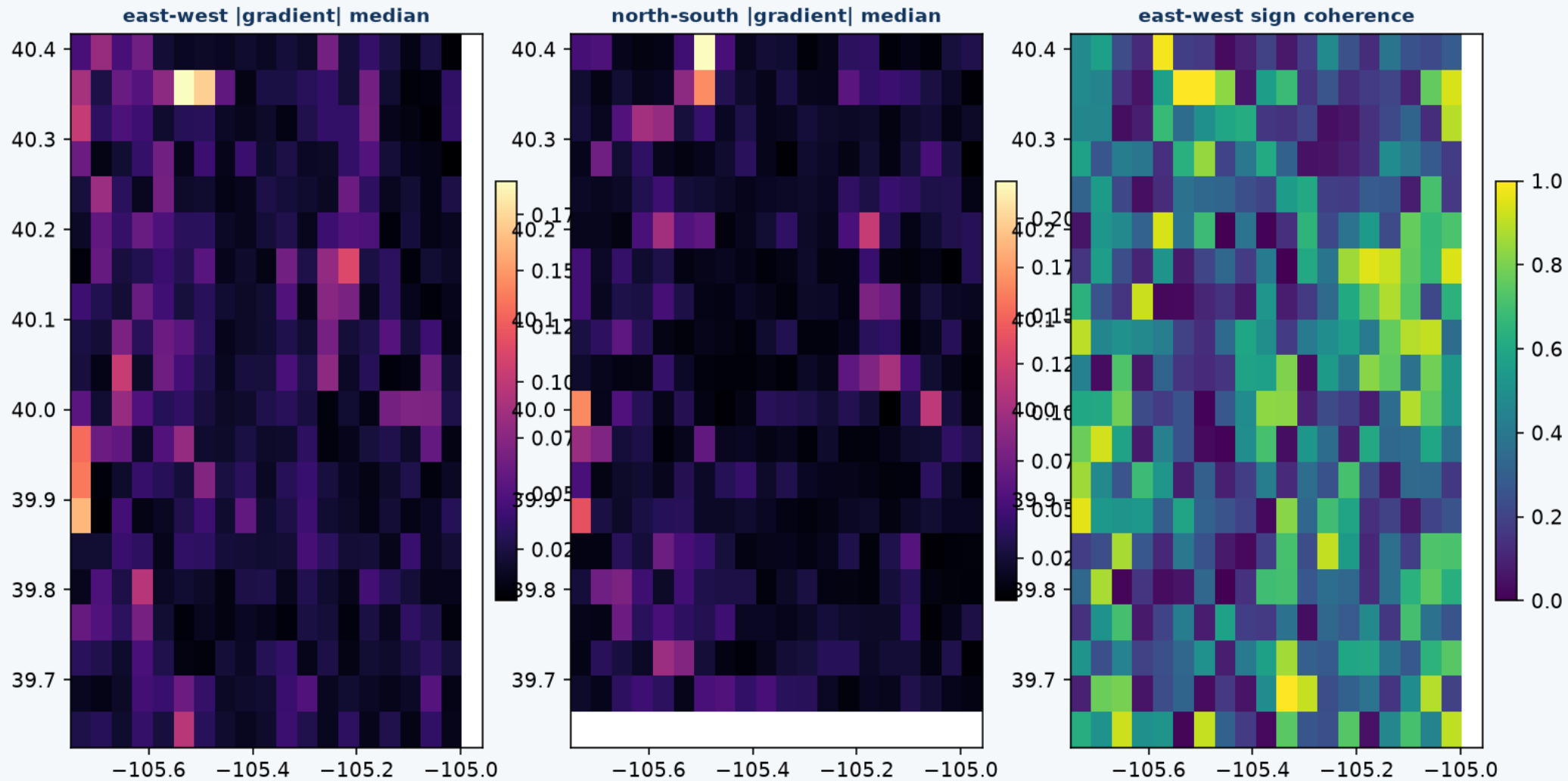


support count



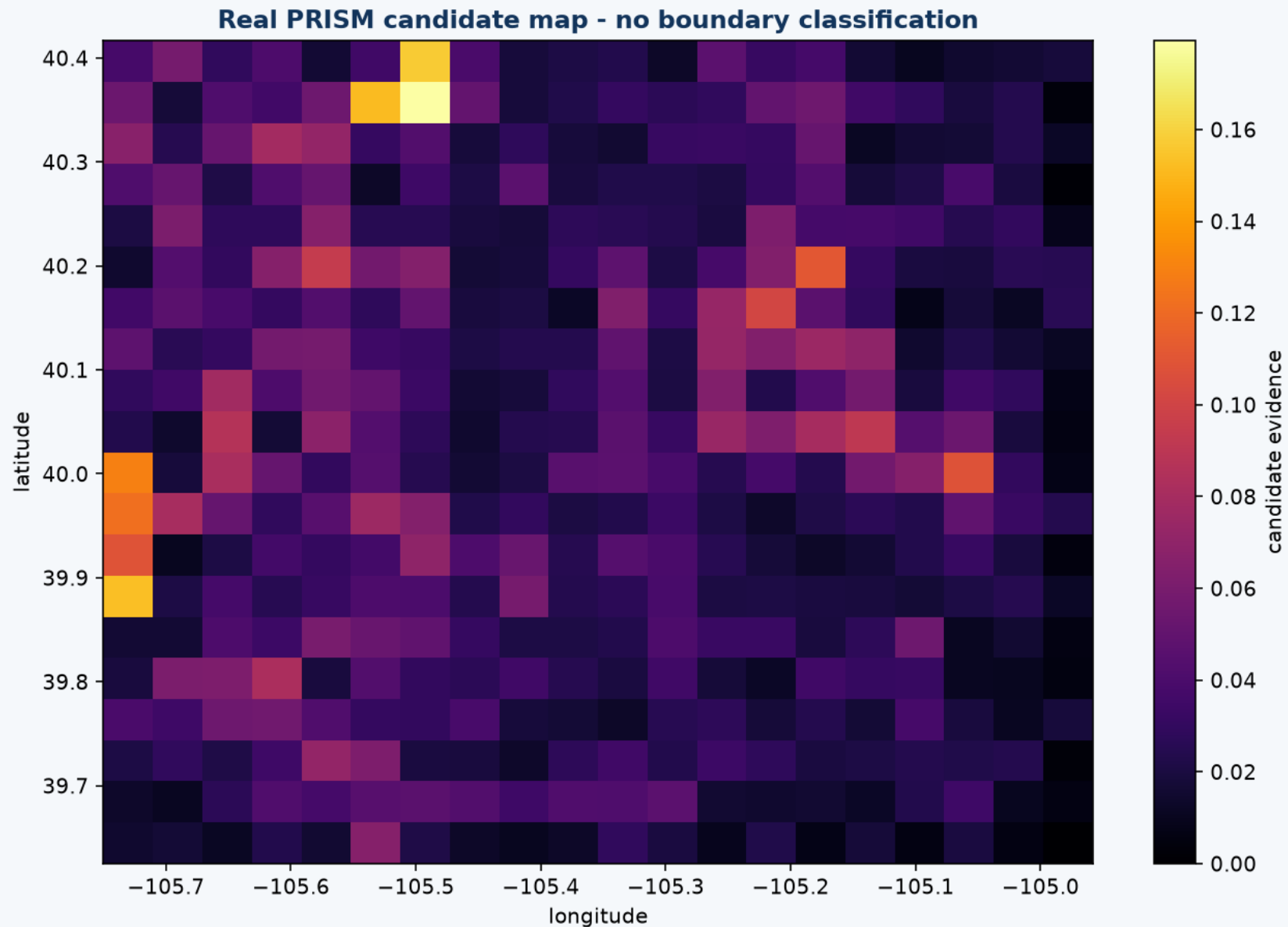
Candidate B - translated-gradient agreement

Different focal surfaces estimate change across the same absolute raster edge



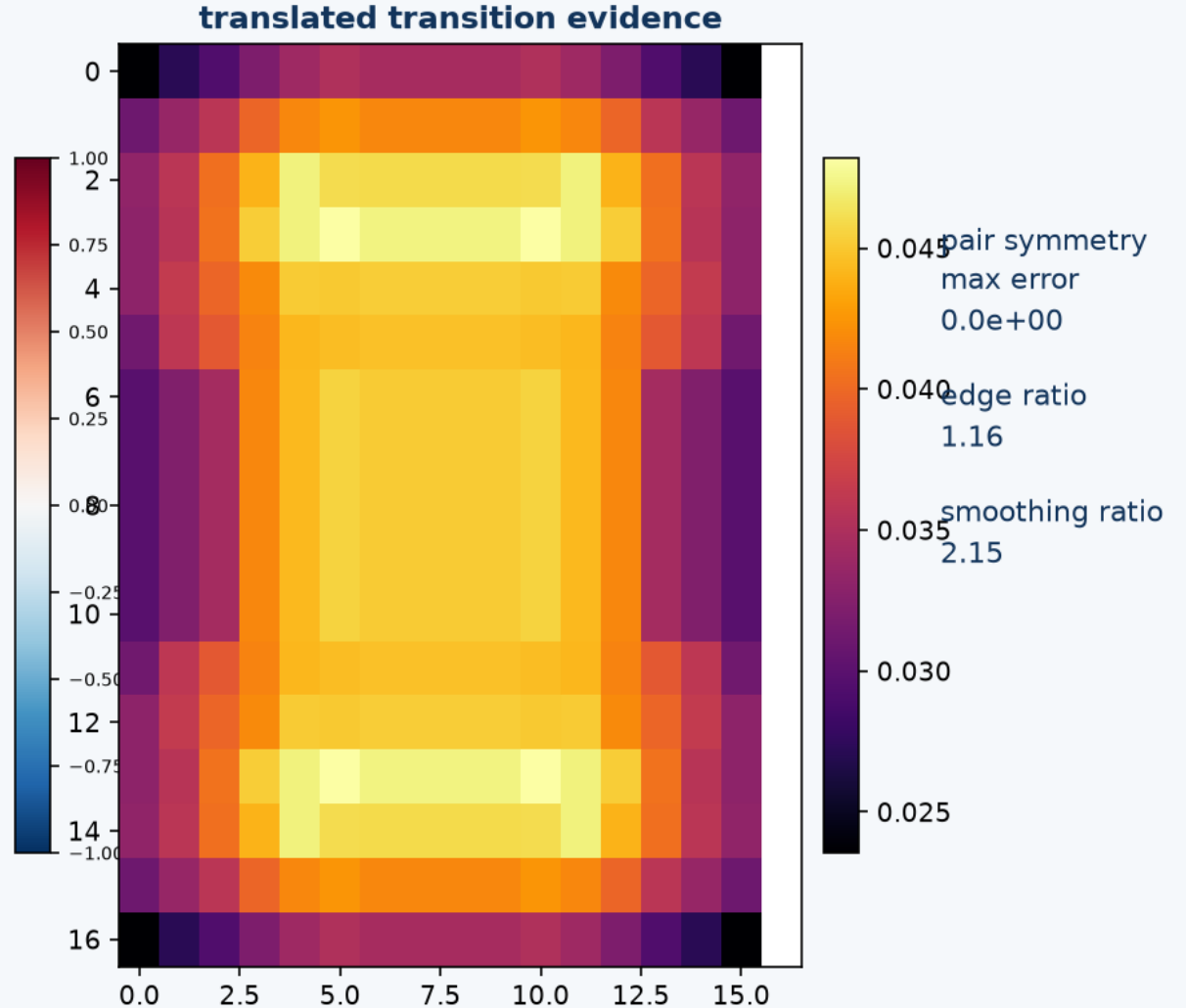
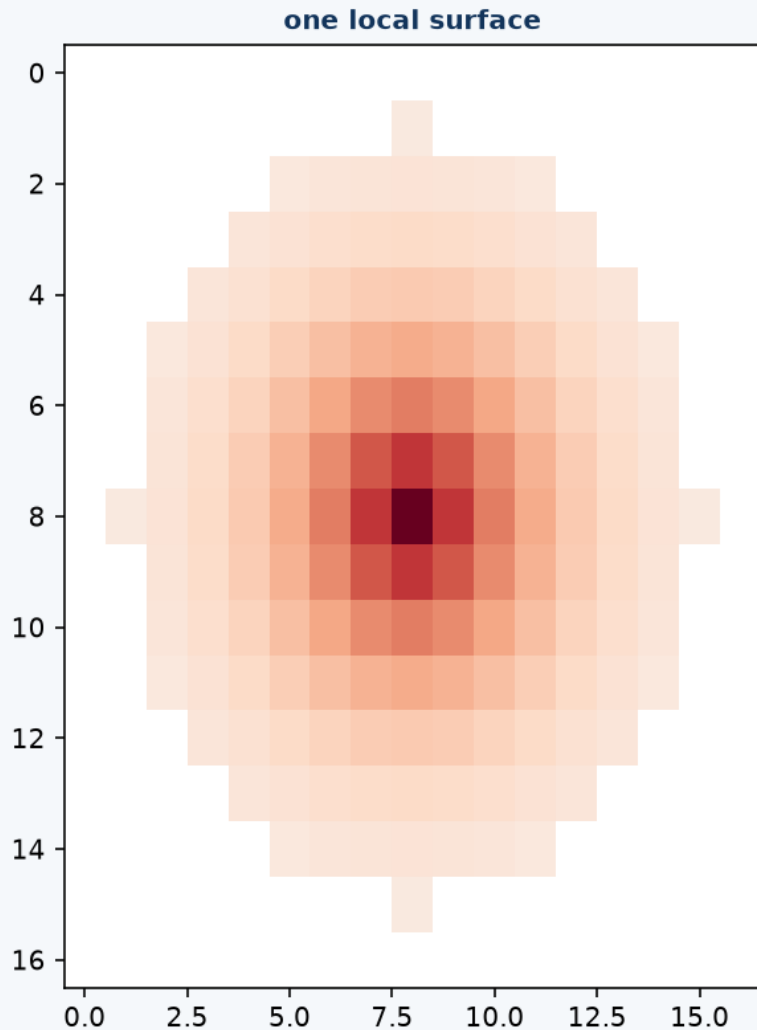
Candidate C - transition evidence is one diagnostic, not the goal

Median absolute aligned gradient weighted by distinct focal support



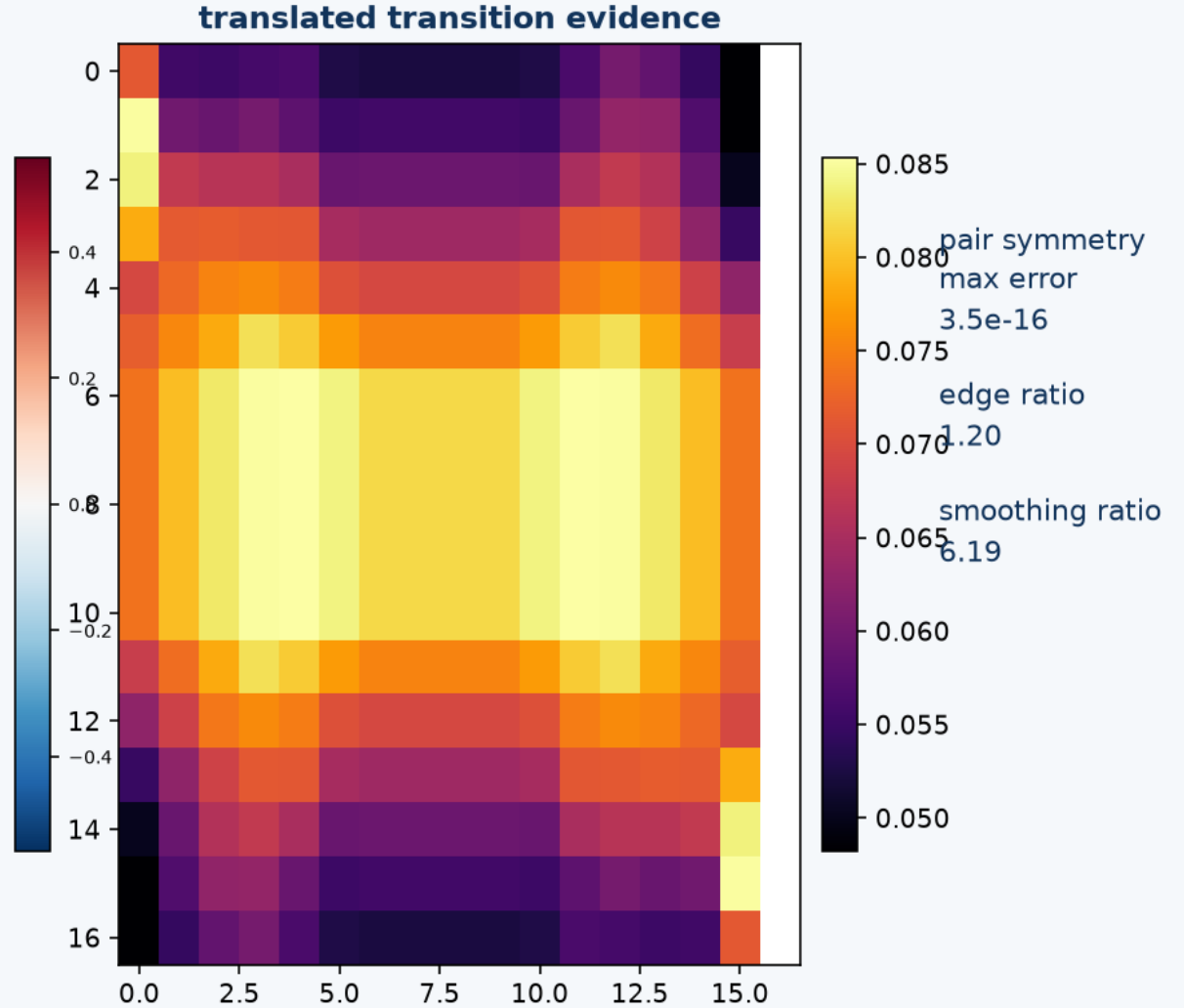
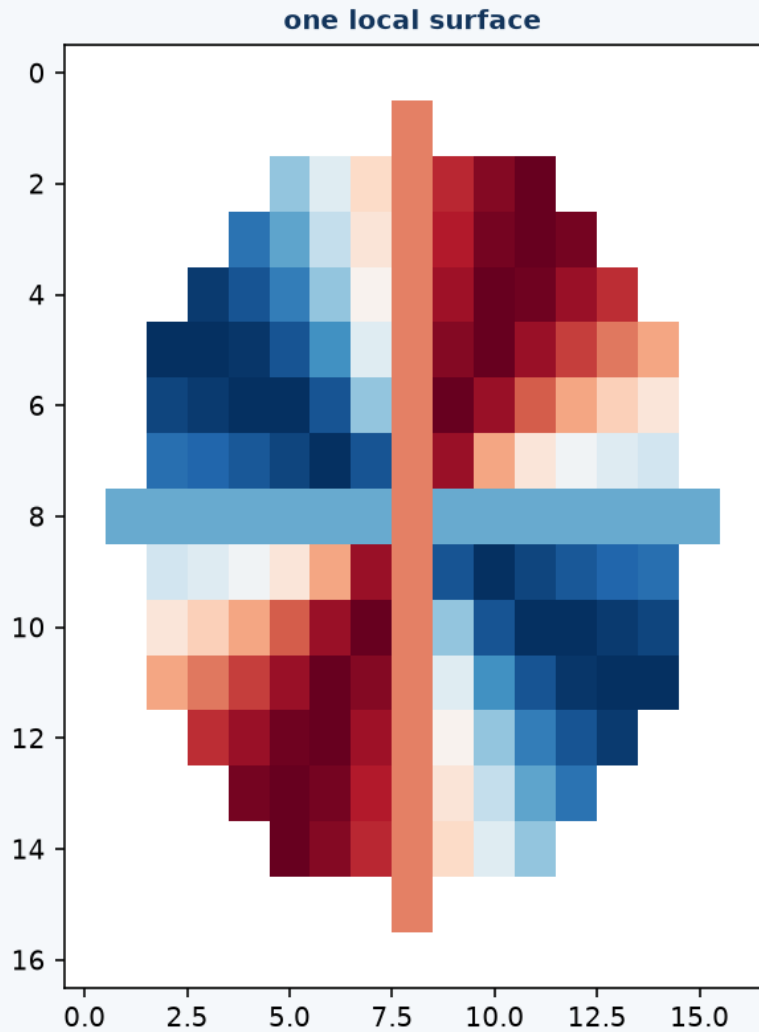
Synthetic isotropic distance decay

Known-answer, pair-symmetric relational field



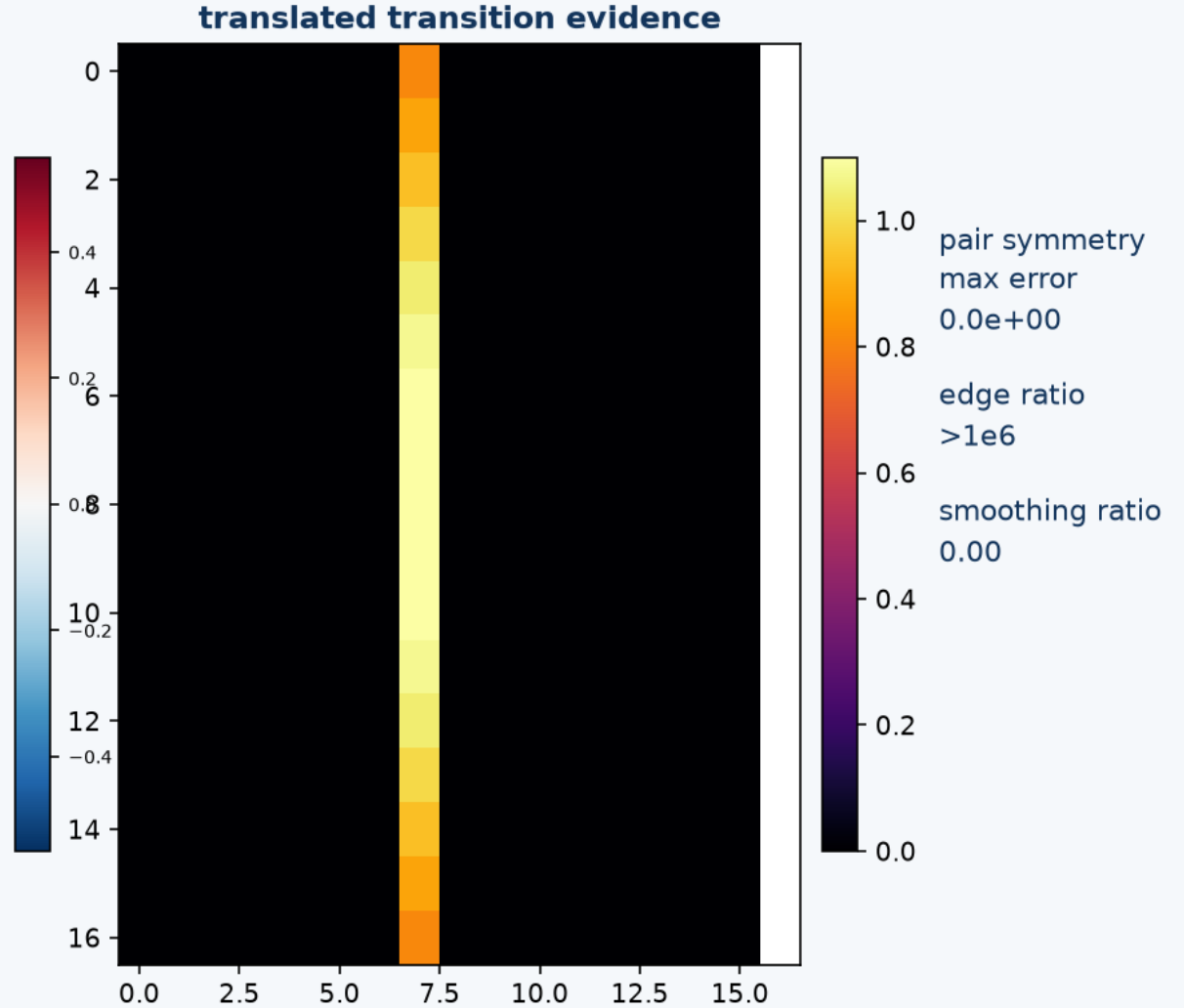
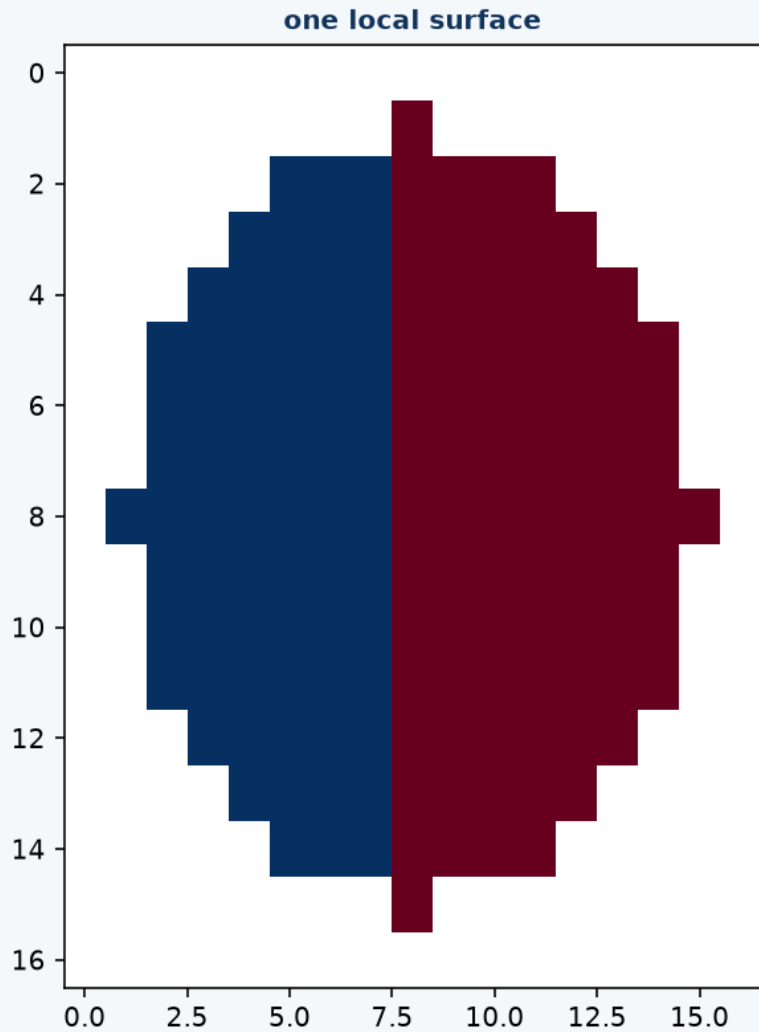
Synthetic directional anisotropy

Known-answer, pair-symmetric relational field



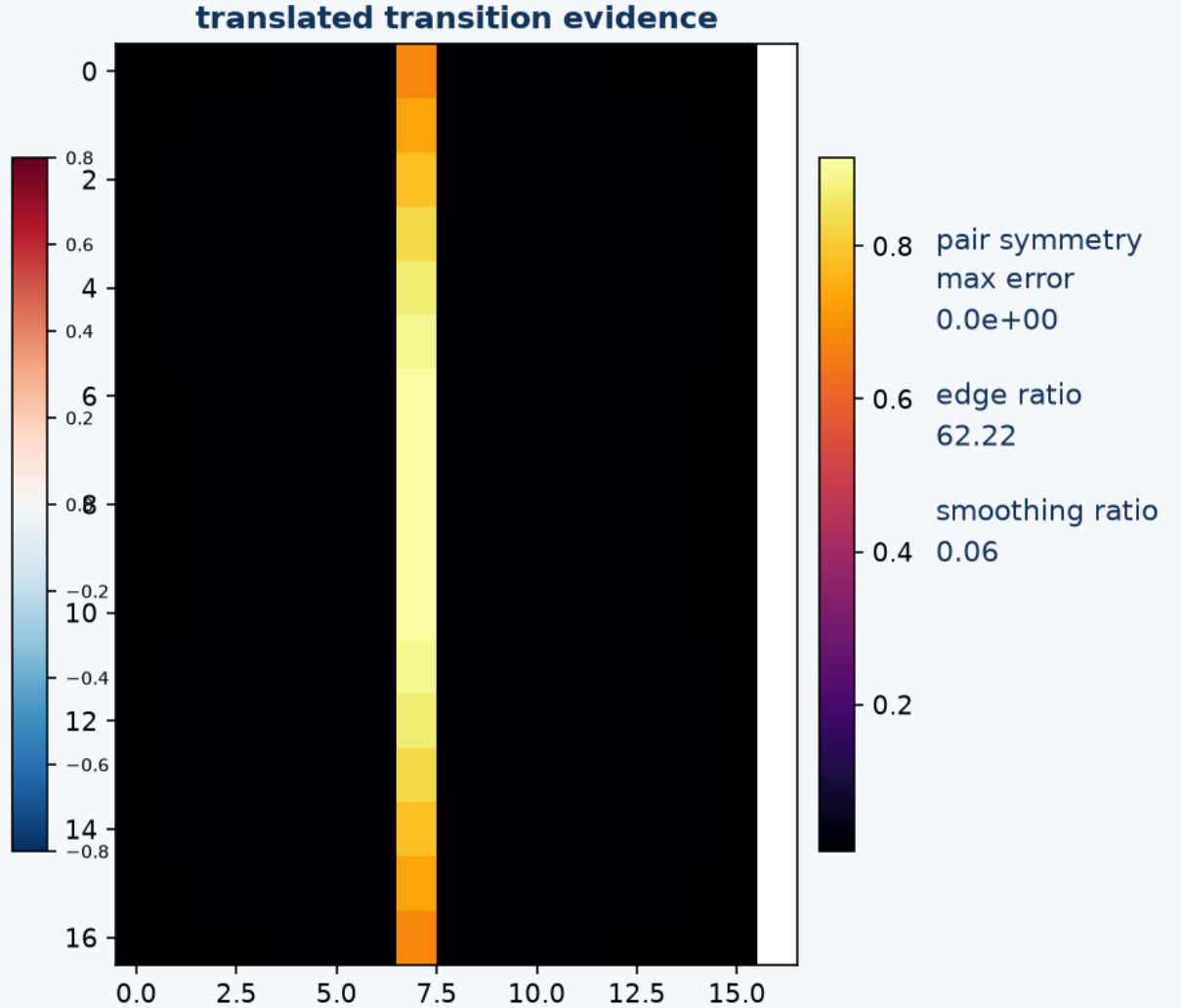
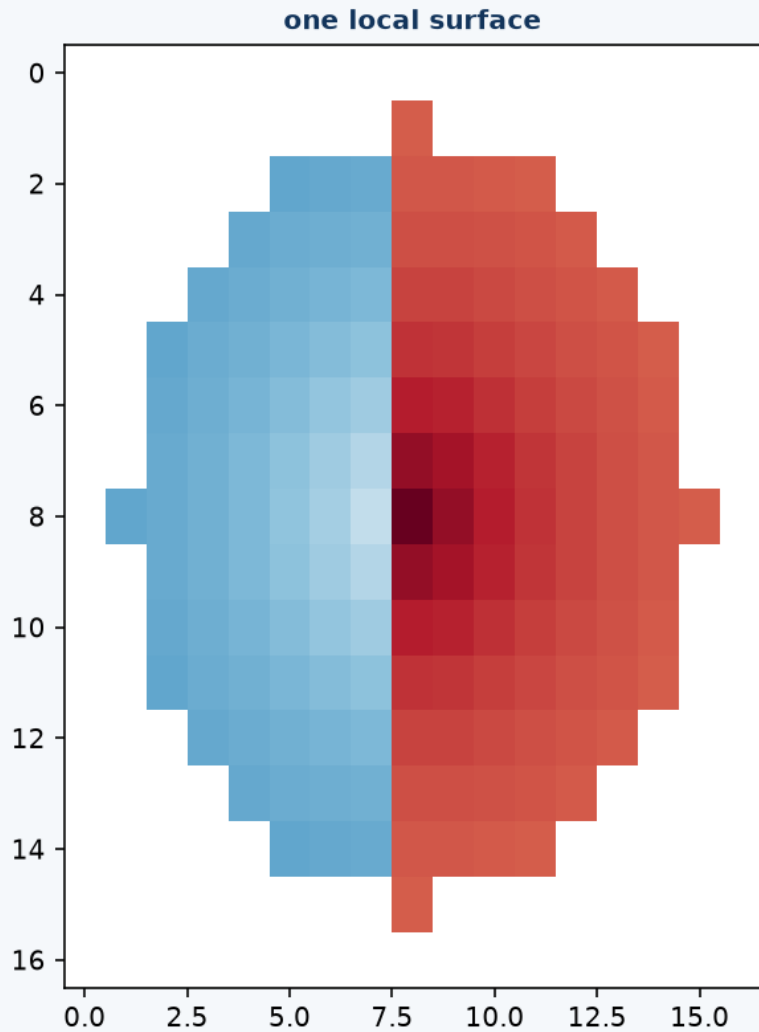
Synthetic sharp transition

Known-answer, pair-symmetric relational field



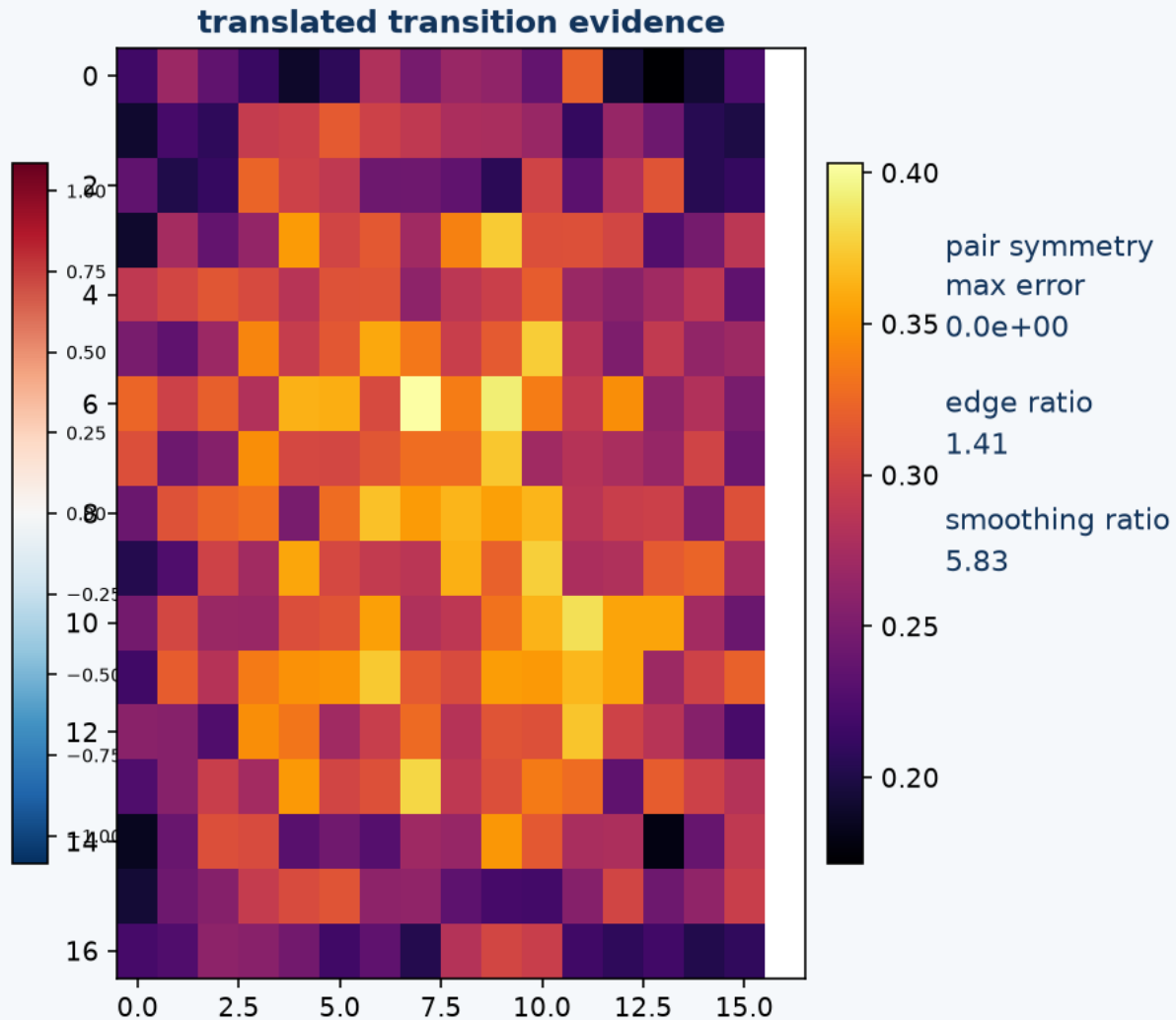
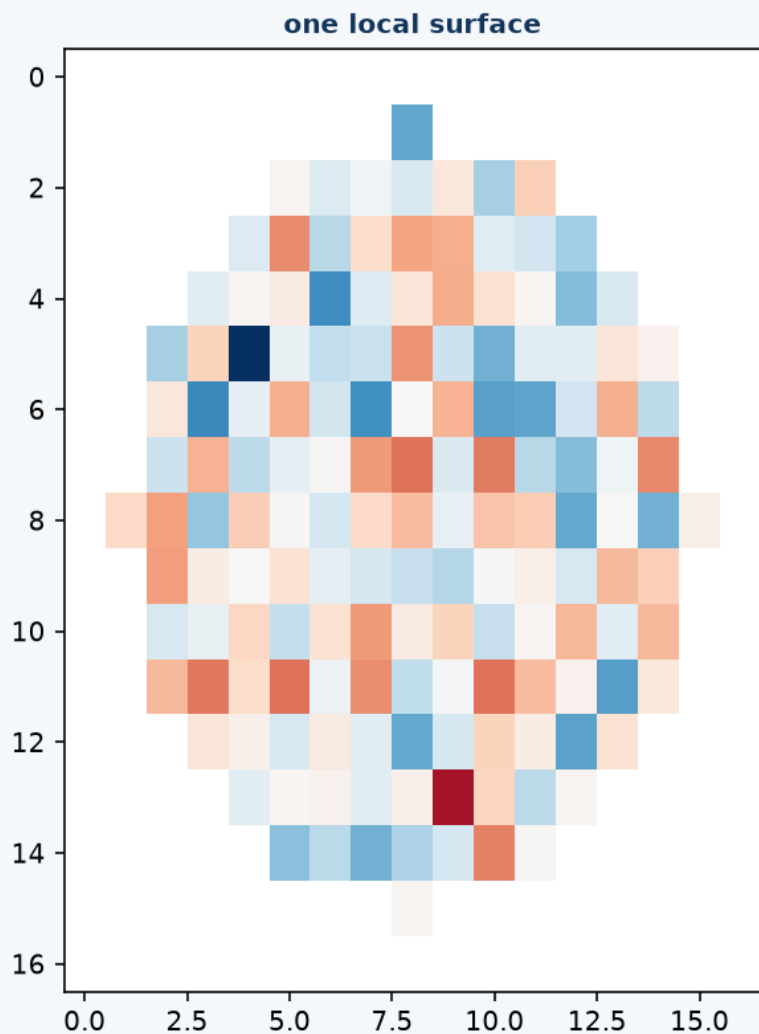
Synthetic transition + distance

Known-answer, pair-symmetric relational field



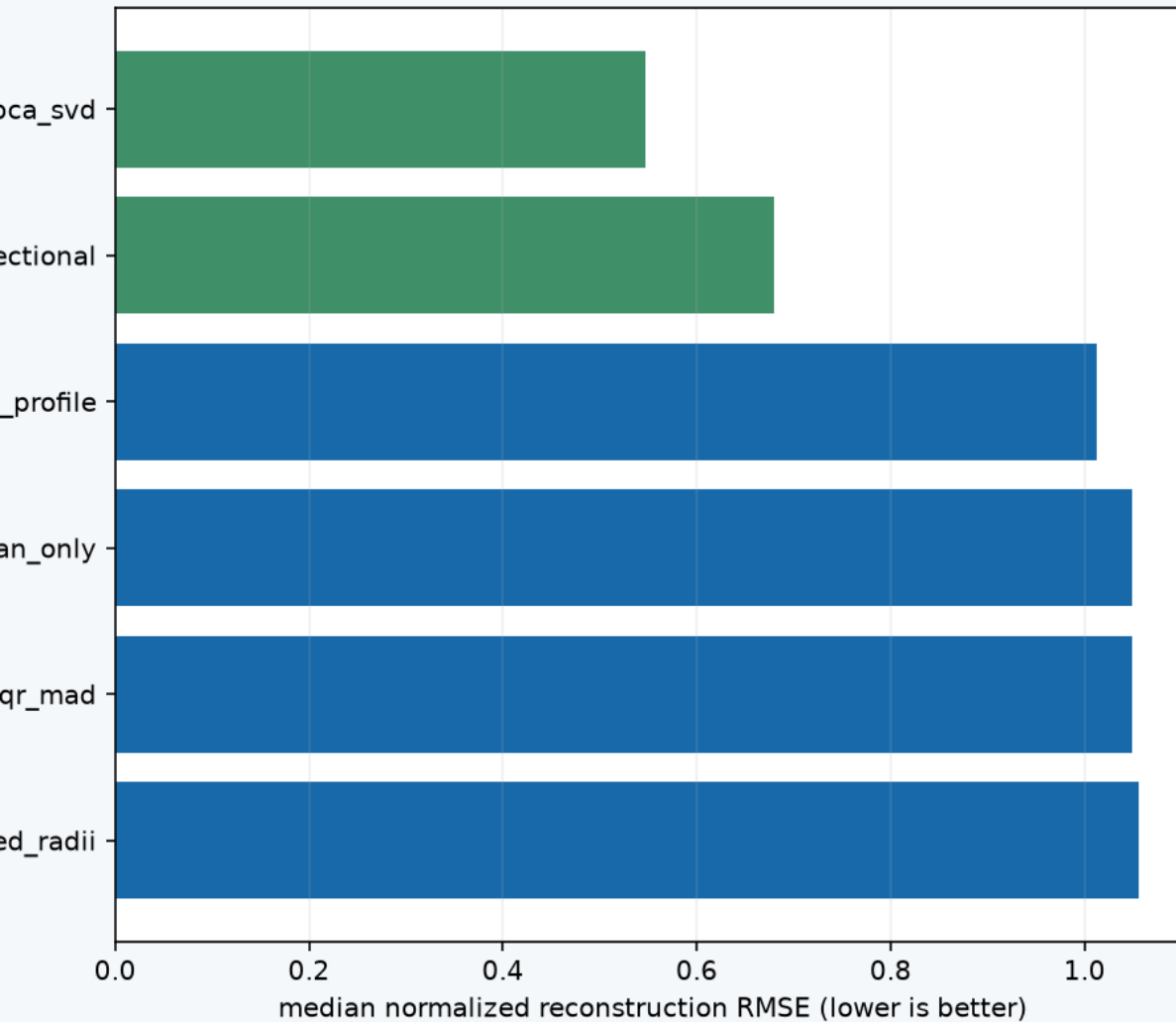
Synthetic noise/null

Known-answer, pair-symmetric relational field



Method comparison: geometry retained versus reconstruction error

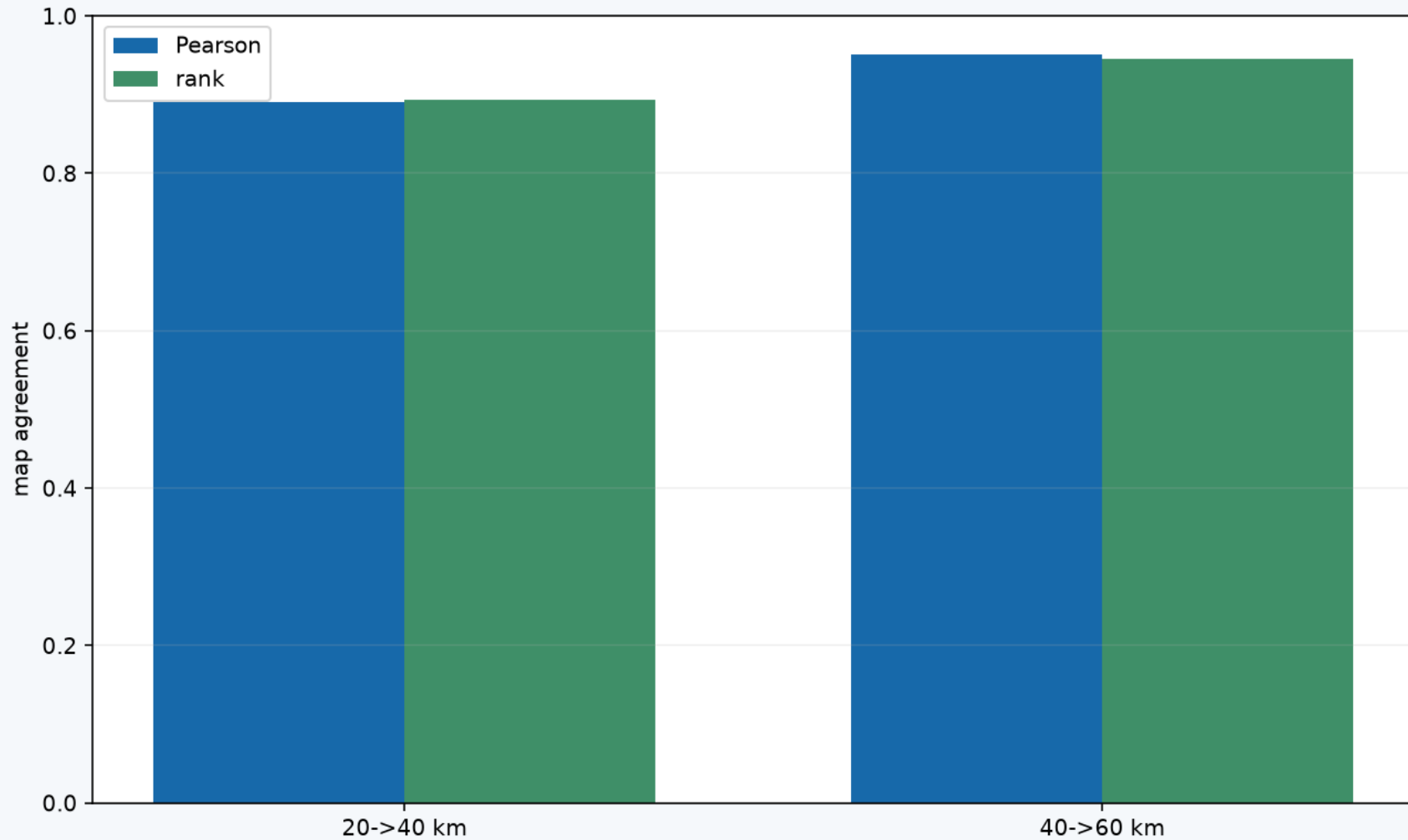
Aesthetics are not a decision criterion



method	recommendation
median only	KEEP AS DIAGNOSTIC
median + IQR/MAD	KEEP AS DIAGNOSTIC
nested radii	KEEP AS DIAGNOSTIC
radial profile	KEEP
radial + directional	KEEP
PCA/SVD	EXPERIMENTAL
overlap coherence	KEEP AS DIAGNOSTIC
translated gradient	EXPERIMENTAL
surface-change operator	KEEP AS DIAGNOSTIC
transition operator	EXPERIMENTAL
learned representation	DROP

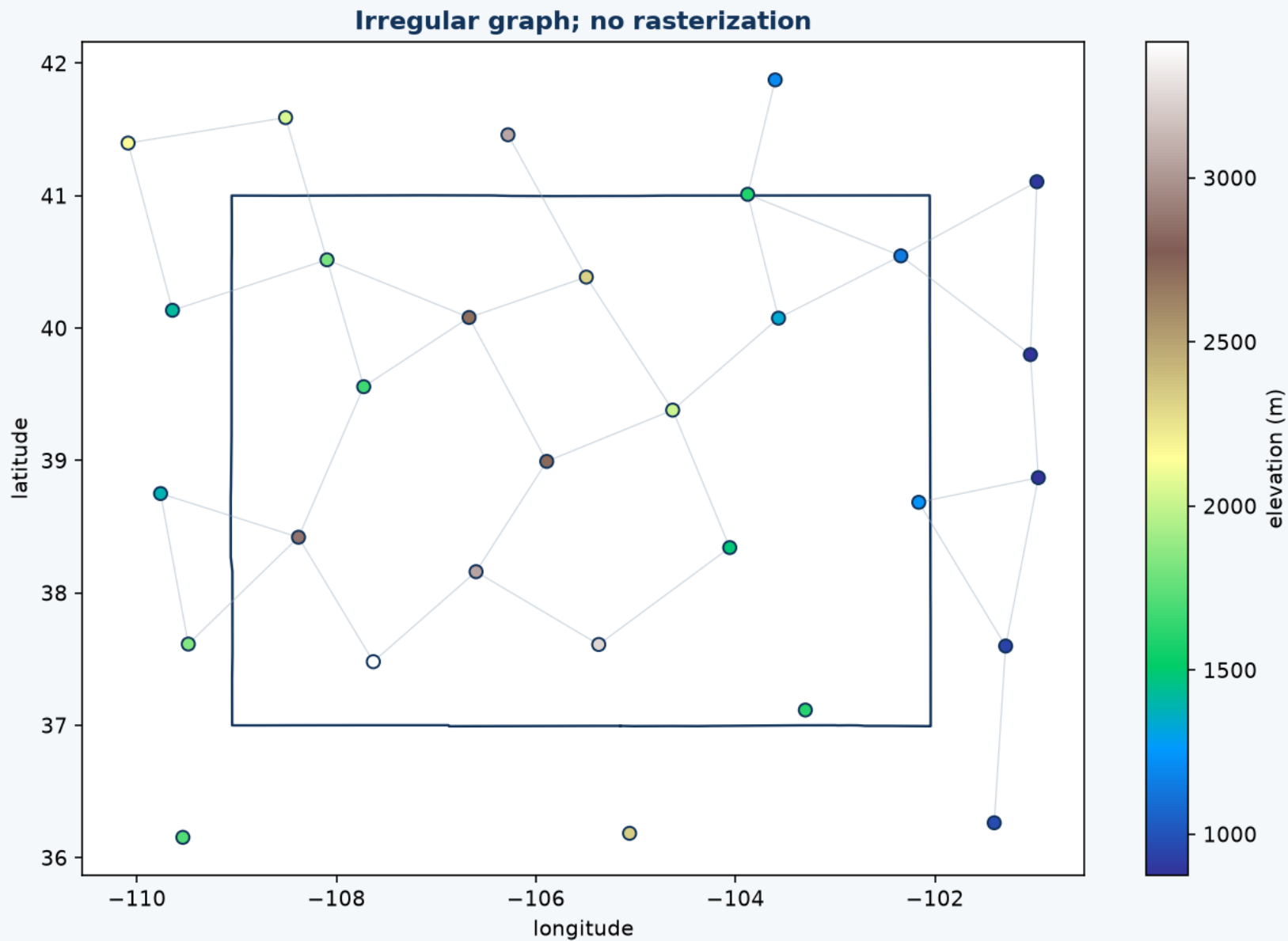
Observation-window stability

Candidate maps persist across practical windows; this does not identify a characteristic scale



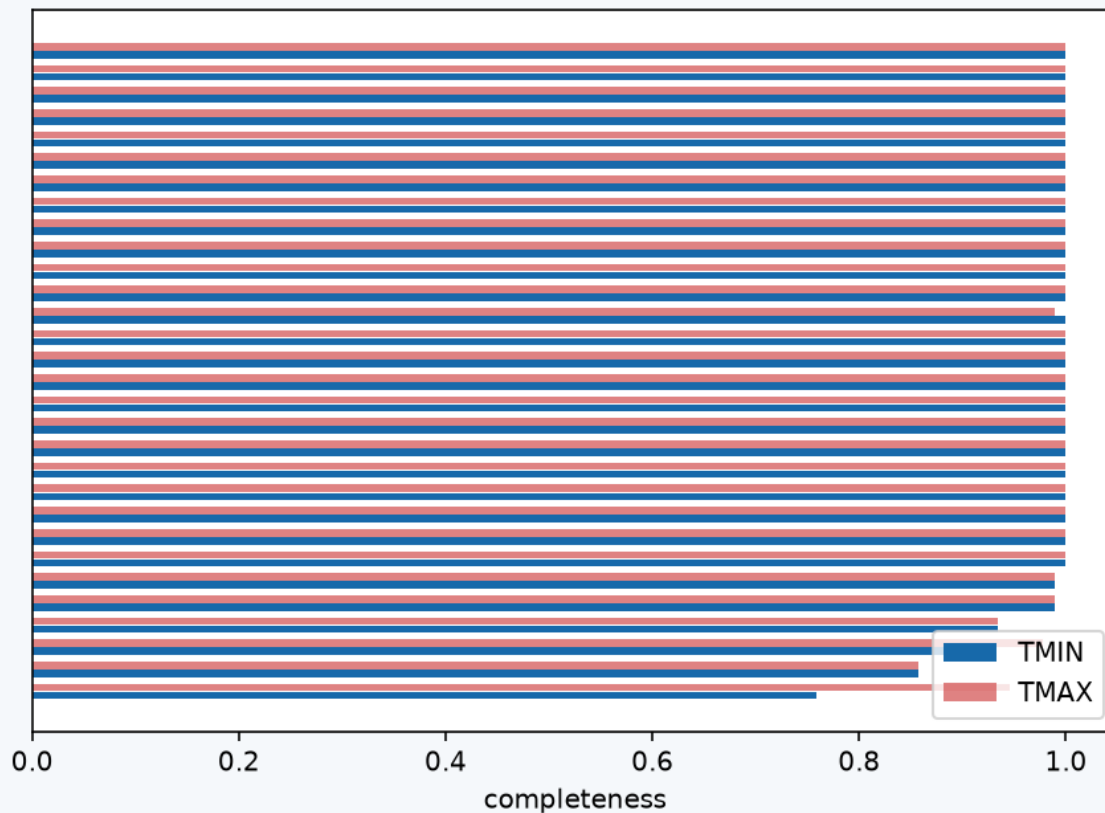
Raw weather-station network

30 selected GHCN-Daily stations; 35 eligible ≤ 150 km edges



Station QC, provenance, and independence limits

Quality-flag exclusions are explicit; PRISM contribution status is not guessed



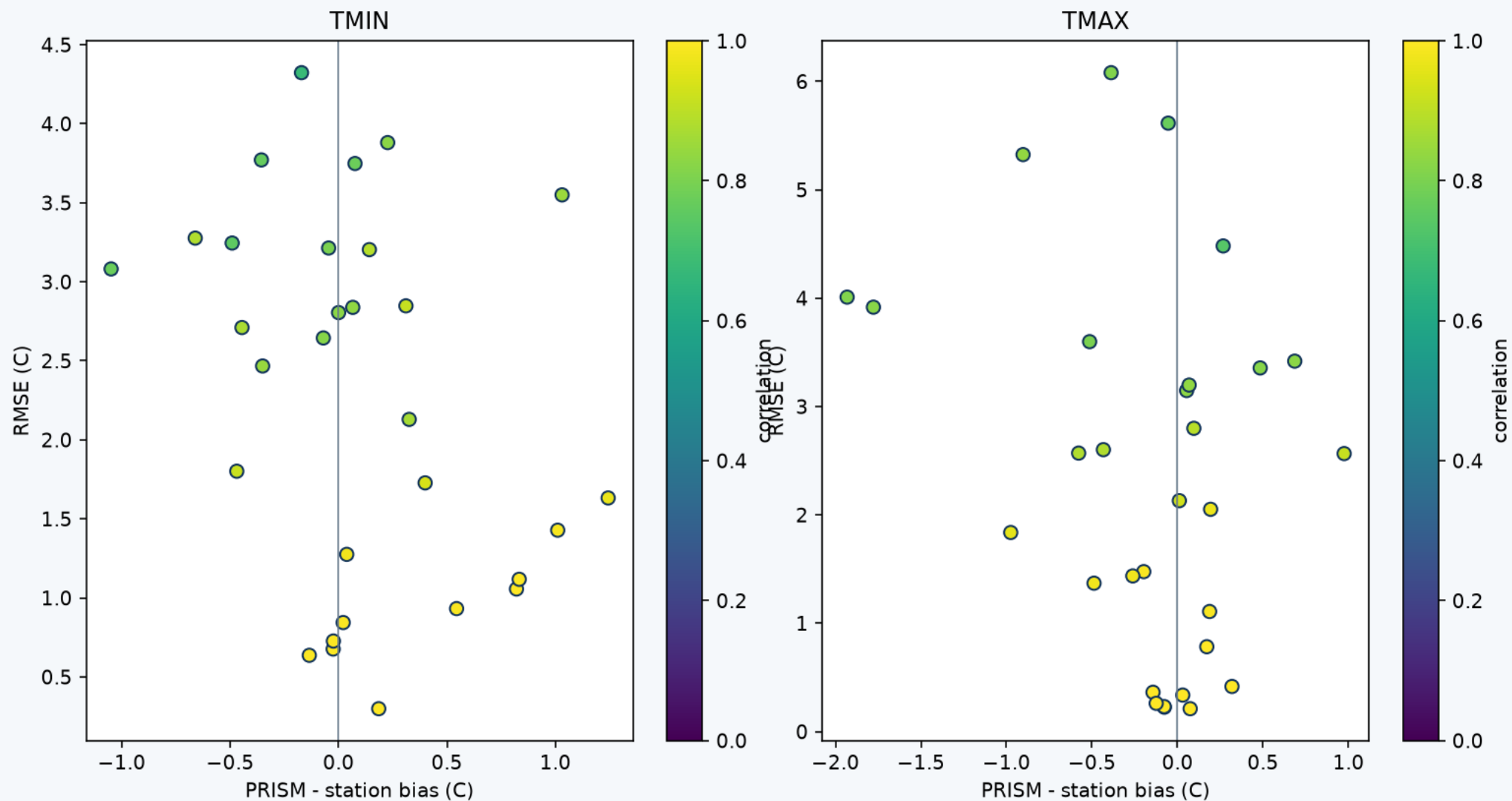
54/60 candidates met the 70% completeness threshold; 30 were selected spatially.

Every selected station is classified UNKNOWN for PRISM contribution status. PRISM AN daily is station-informed, and the exact product-period roster was not established.

PRISM day: 1200-1200 UTC, day-ending label. GHCN observation time is local when present. Same-label and +/-1-day sensitivity are retained.

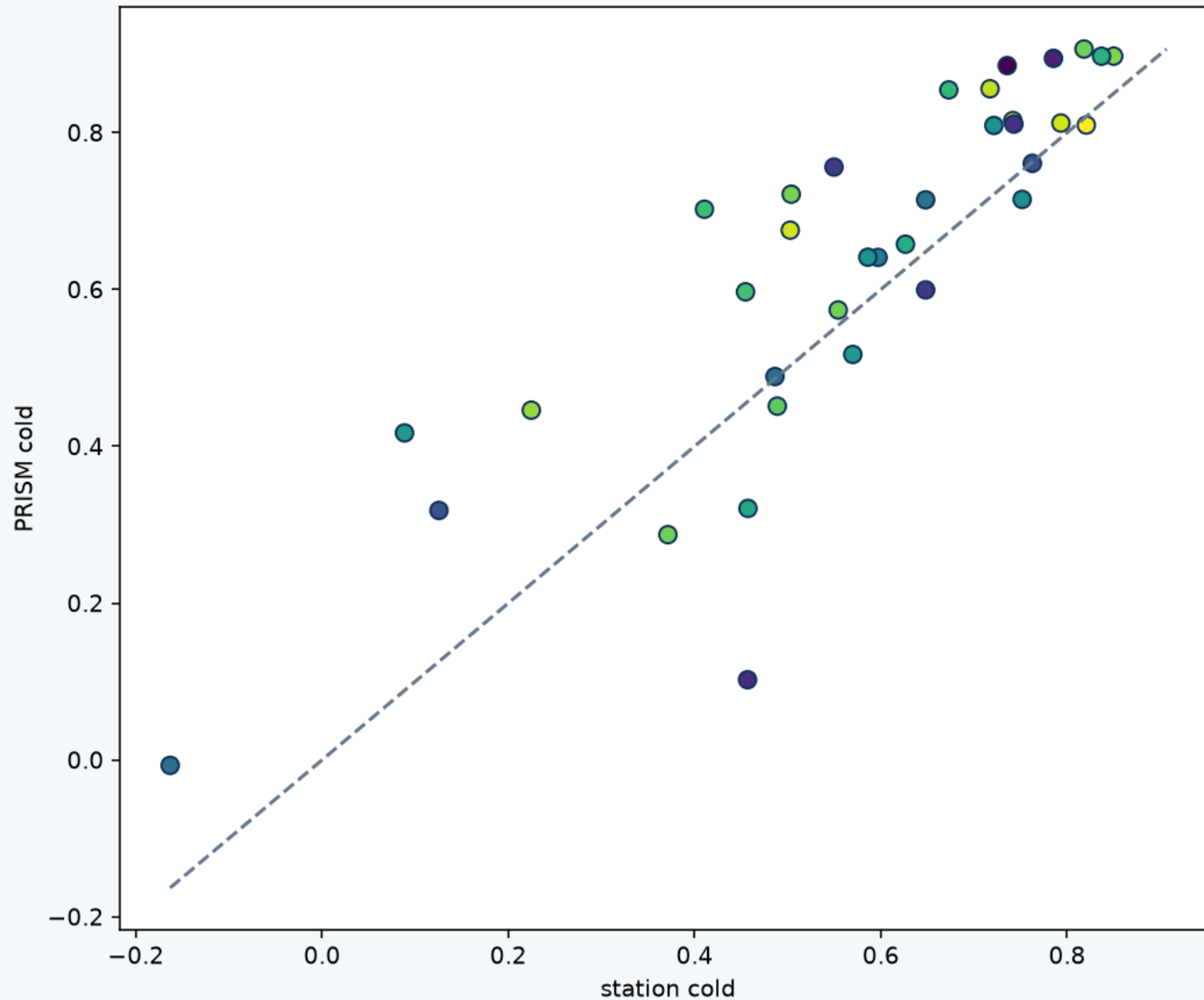
Raw station versus PRISM TMIN/TMAX

Station-level same-label metrics; +/-1-day comparisons remain in the artifact table



Station versus PRISM cold pair synchrony

Station thresholds come from station observations; PRISM thresholds come from PRISM

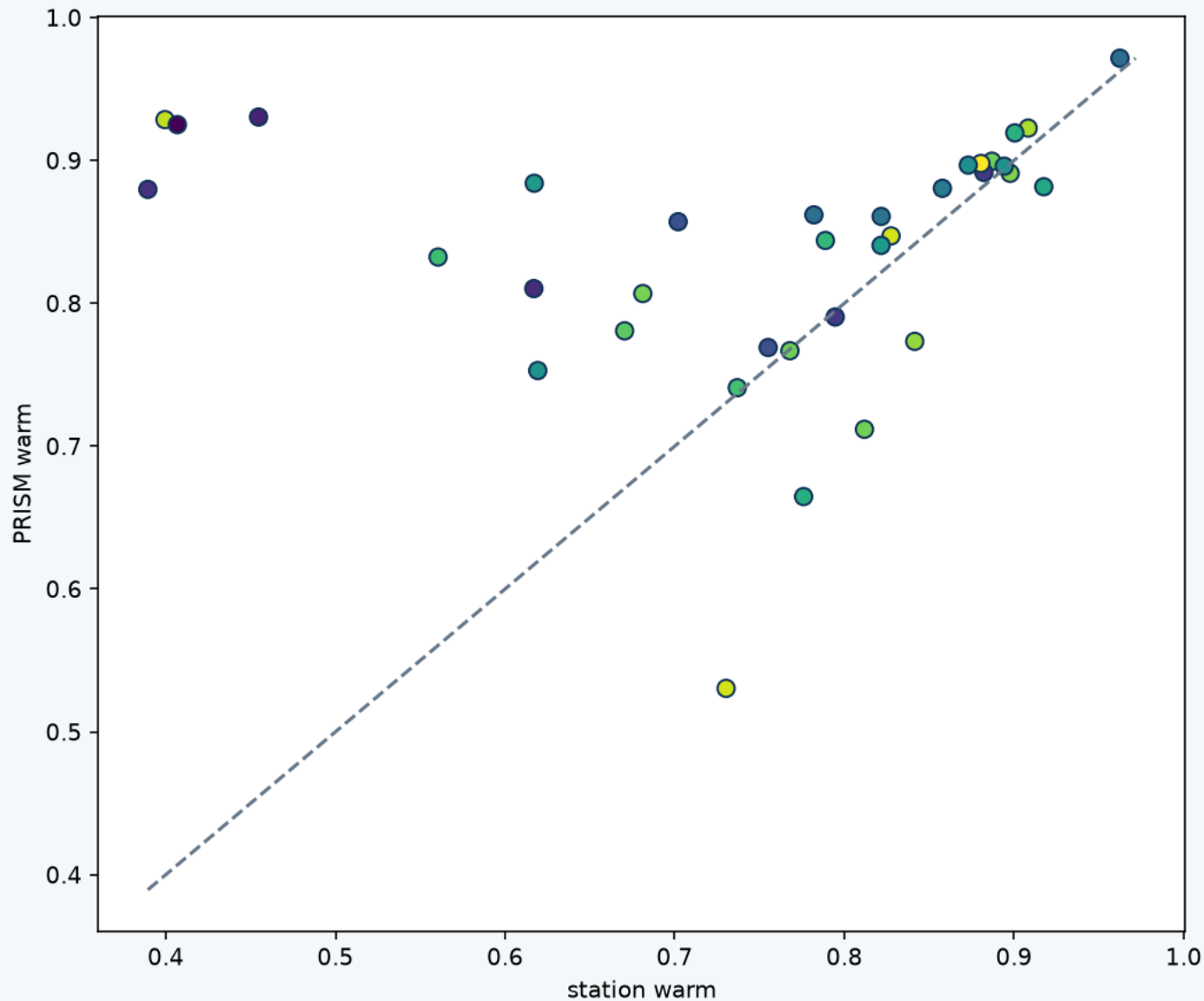


n edges = 35 $r = 0.84$ rank $r = 0.87$ MAE = 0.11 RMSE = 0.14

Edges share stations. These are descriptive, dependence-aware in wording, and not independent validation.

Station versus PRISM warm pair synchrony

Station thresholds come from station observations; PRISM thresholds come from PRISM

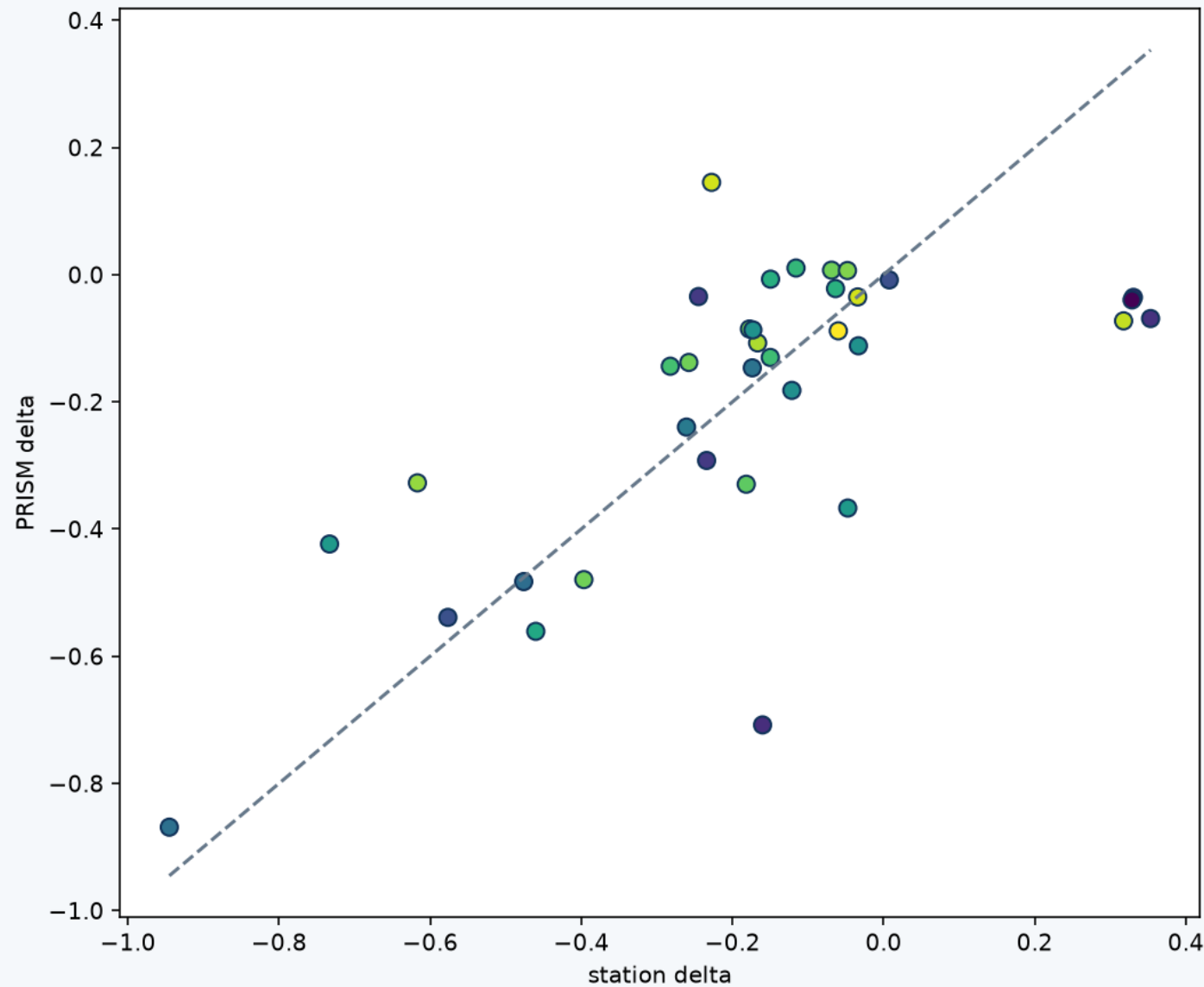


n edges = 35 $r = 0.03$ rank $r = 0.30$ MAE = 0.12 RMSE = 0.20

Edges share stations. These are descriptive, dependence-aware in wording, and not independent validation.

Station versus PRISM delta pair synchrony

Station thresholds come from station observations; PRISM thresholds come from PRISM

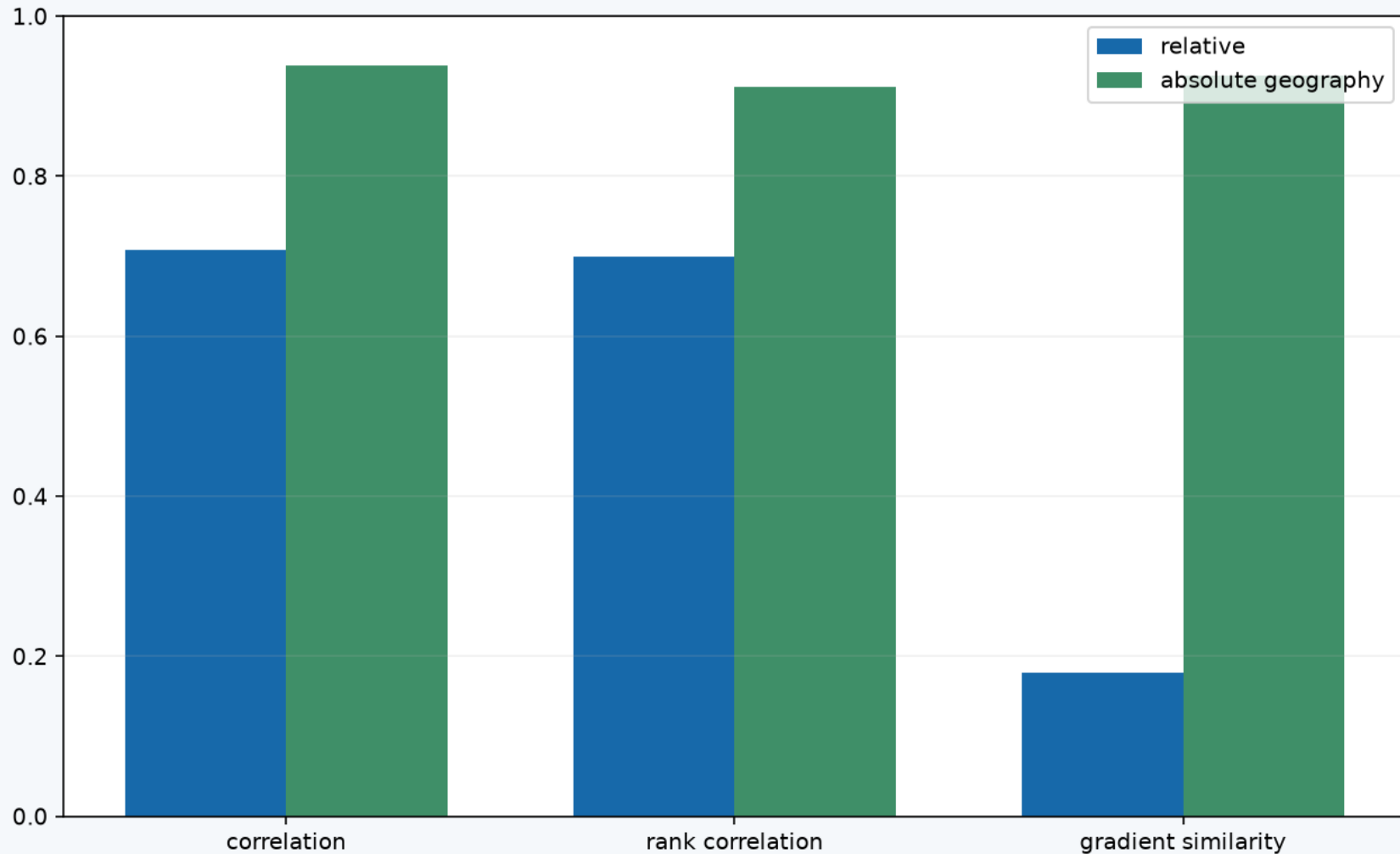


n edges = 35 $r = 0.68$ rank $r = 0.62$ MAE = 0.15 RMSE = 0.21

Edges share stations. These are descriptive, dependence-aware in wording, and not independent validation.

What overlap adds in the real block

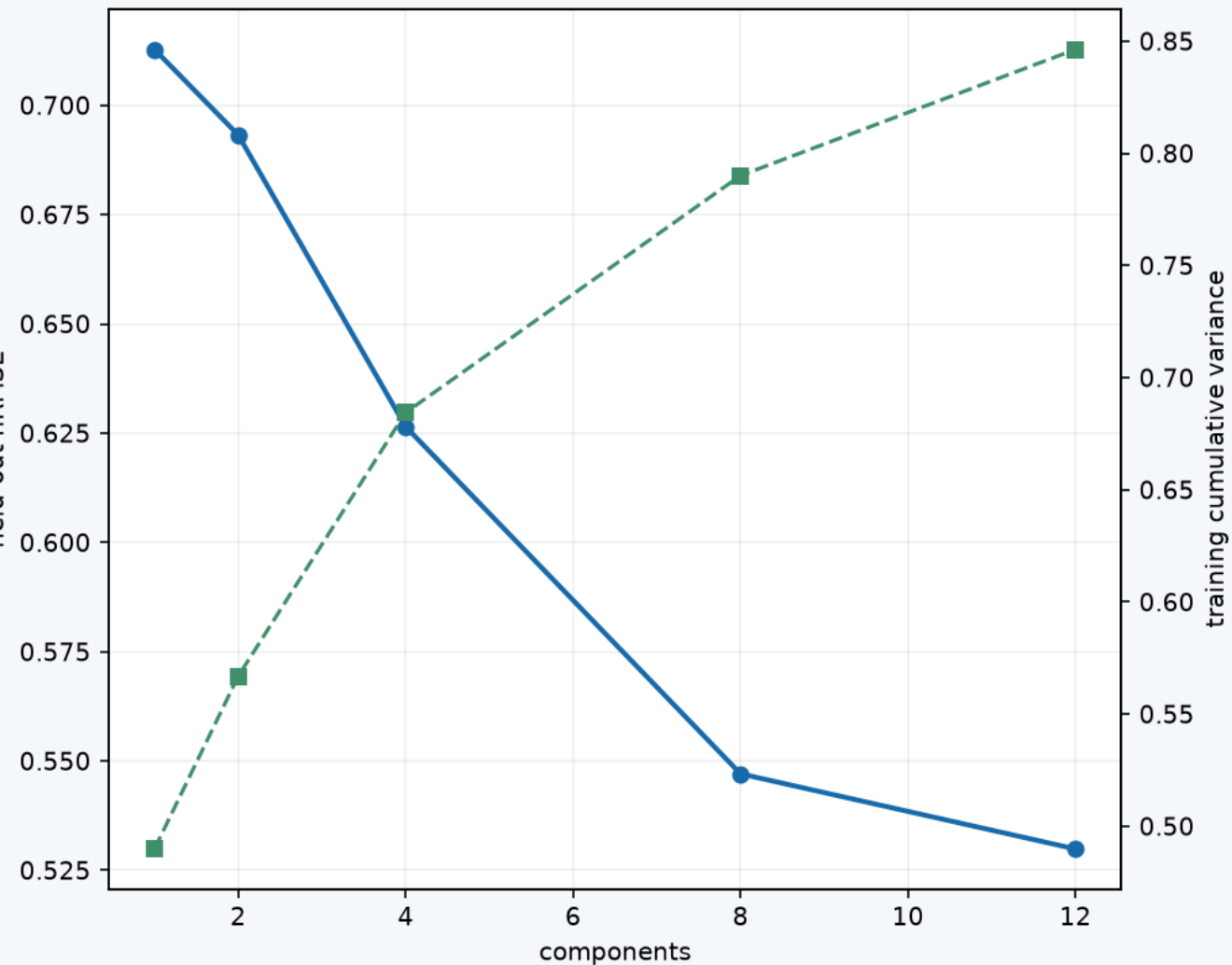
Absolute alignment uses known geography and outperforms relative-offset comparison



This gain is not created by pair reversal: canonical identities are explicit, and edge support excludes focal-self gradients.

PCA/SVD is useful, but no autoencoder is justified

Held-out reconstruction improves with components; transparency wins for this feasibility decision



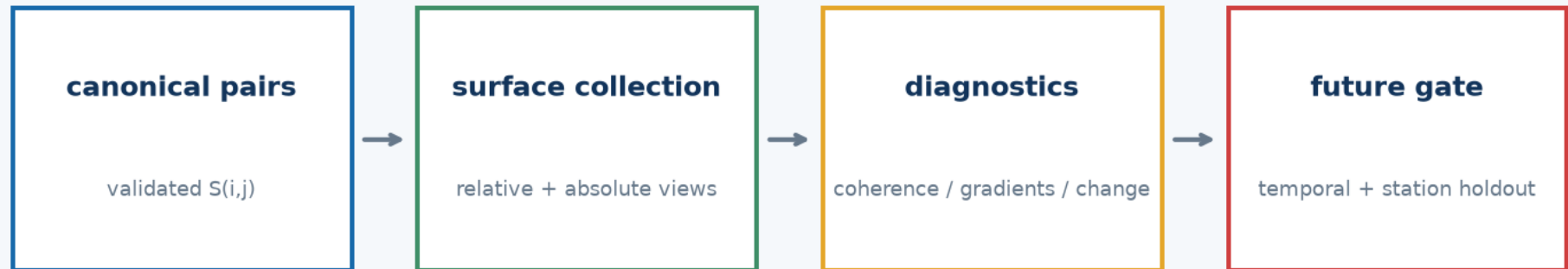
8-component held-out nRMSE: 0.55 on 64 interior surfaces.

Radial + directional nRMSE: 0.68, with direct scientific interpretation.

A small autoencoder was not run: it would add dependency and opacity without changing today's decision.

Recommendation: preserve relations, add an experimental aligned-view layer

Stop here - no Colorado second-stage production run and no new public verb



WORKED - Known geographic alignment, translated gradients, and explicit support reveal stable structure beyond independent reductions and smoothing controls.

WEAK - Median/radius reductions lose geometry; warm station-pair agreement is weak; the station graph is sparse and not demonstrably independent of PRISM.

NEXT - Keep an internal canonical-pair-backed surface collection with reversible relative/absolute views. Require temporal robustness and station-level/spatial-block holdouts before promotion.